

P25 J3398: Non-uniform Refinement

CryoSPARC Running Version: v4.7.1

Project Details

DETAIL	VALUE
Title	24sep20b_WT-Control-White_From-January
Description	This is a repeat collection of White Control Wildtype CFTR from the original January/February 2024 collection. Using
ID	P25
Created by user	Andy Paige

Job Details

DETAIL	VALUE
Title	New Job J3398
Description	Enter a description.
ID	J3398
Job type	Non-uniform Refinement
Status	completed
Created by user	Minou Emmad
Created	Tue Jun 09 2026 15:14:14 GMT-0400 (Eastern Daylight Time)
Queued	Tue Jun 09 2026 15:14:47 GMT-0400 (Eastern Daylight Time)
Launched	Tue Jun 09 2026 15:16:56 GMT-0400 (Eastern Daylight Time)
Started	Tue Jun 09 2026 15:17:04 GMT-0400 (Eastern Daylight Time)
Waiting	-
Killed	-
Completed	Tue Jun 09 2026 15:28:45 GMT-0400 (Eastern Daylight Time)
Failed	-
Last accessed	Wed Jun 10 2026 15:55:07 GMT-0400 (Eastern Daylight Time)
Size	3.55 GB

Inputs

particles

J3393.particles_all_classes

BLOB
J3393.particles_all_classes.blob.F
CTF
J3393.particles_all_classes.ctf.F

ALIGNMENTS3D

J3393.particles_all_classes.alignments3D.F
PASSTHROUGH
J3393.particles_all_classes.alignments3D_multi.F

volume

J3393.volume_class_0
MAP
J3393.volume_class_0.map.F

mask

No group connections

Parameters

Particle preprocessing

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED
Window dataset (real-space)	True	X		
Window inner radius	0.85	X		X
Window outer radius	0.99	X		X

Homogeneous Refinement

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED
Symmetry	C1	X		
Symmetry relaxation method	none	X		X
Do symmetry alignment	True	X		X
Re-estimate greyscale level of input reference	True	X		X
Number of extra final passes	0	X		X
Maximum align resolution (A)	None	X		
Initial lowpass resolution (A)	30	X		X
GSFSC split resolution (A)	20	X		X
Force re-do GS split	False	X		X
Adaptive Marginalization	True	X		X
Non-uniform refine enable	True	X		X
Non-uniform filter order	8	X		X
Non-uniform AWF	3	X		X
Enforce non-negativity	False	X		X
Window structure in real space	True	X		X
Skip interpolant premult	True	X		X
Ignore DC component	True	X		X
Ignore tilt	False	X		X

Ignore trefoil	False	X		X	
Ignore tetra	False	X		X	
Use random ordering of particles	True	X		X	
Disable auto batchsize	False	X		X	
Batchsize epsilon	0.001	X		X	
Batchsize snrfactor	50	X		X	
Reset input per-particle scale	True	X		X	
Minimize over per-particle scale	False	X		X	
Scale min/use start iter	0	X		X	
Noise priorw	50	X		X	
Noise initw	200	X		X	
Noise initial sigma-scale	3	X		X	
Initialize noise model from images	False	X		X	
Dynamic mask threshold (0-1)	0.2	X		X	
Dynamic mask near (A)	6	X		X	
Dynamic mask far (A)	14	X		X	
Dynamic mask start resolution (A)	12	X			
Dynamic mask use absolute value	False	X		X	
Show plots from intermediate steps	True	X			
GPU batch size of images	None	X		X	

Defocus Refinement

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Optimize per-particle defocus	False	X		X	
Num. particles to plot	3	X			
Minimum Fit Res (A)	20	X			
Defocus Search Range (A +/-)	2000	X			
GPU batch size of images	None	X		X	

Global CTF Refinement

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Optimize per-group CTF params	False	X		X	
Num. groups to plot	3	X			
Binning to apply to plots	1	X		X	
Minimum Fit Res (A)	10	X			
Fit Tilt	True	X			
Fit Trefoil	True	X			

Fit Spherical Aberration	False	X			
Fit Tetrafoil	False	X			
Fit Anisotropic Mag.	False	X			
GPU batch size of images	None	X		X	

Ewald Sphere Correction

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Do EWS correction	False	X		X	
EWS curvature sign	negative	X		X	

Random Seeds

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Random seed	57167689	X		X	

Compute settings

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Cache particle images on SSD	True	X			
Low-Memory Mode	False	X			

Outputs

particles

ALIGNMENTS3D	
J3398.particle.alignments3D	

CTF	
J3398.particle.ctf	

BLOB (PASSTHROUGH)	
J3398.particle.blob	

ALIGNMENTS3D_MULTI (PASSTHROUGH)	
J3398.particle.alignments3D_multi	

volume

MAP	
J3398.volume.blob	

MAP_SHARP	
J3398.volume.blob	

MAP_HALF_A	

J3398.volume.blob	
MAP_HALF_B	
J3398.volume.blob	
MASK_REFINE	
J3398.volume.blob	
MASK_FSC	
J3398.volume.blob	
MASK_FSC_AUTO	
J3398.volume.blob	
PRECISION	
J3398.volume.blob	

mask

MASK_REFINE	
J3398.volume.blob	

Target

ganges.biology.columbia.edu (node)

```
> [2026-06-09 15:16:56.46 GMT-4] License is valid.
> [2026-06-09 15:16:56.46 GMT-4] Launching job on lane default target ganges.biology.columbia.edu ...
> [2026-06-09 15:16:56.48 GMT-4] Running job on master node hostname ganges.biology.columbia.edu
> [2026-06-09 15:17:04.09 GMT-4] [CPU: 78.9 MB]      [Avail: 317.71 GB] Job J3398 Started
> [2026-06-09 15:17:04.11 GMT-4] [CPU: 78.9 MB]      [Avail: 317.71 GB] Master running v4.7.1, worker running v4.7.1
> [2026-06-09 15:17:04.12 GMT-4] [CPU: 79.2 MB]      [Avail: 317.71 GB] Working in directory:
                                /mnt/nfs_HuntLabSynologyDS1621CryoEM/hCFTR/Movies/24sep20b/24
                                sep20b/rawdata/CS-24sep20b-wt-control-white-from-january/J339
                                8
> [2026-06-09 15:17:04.13 GMT-4] [CPU: 79.2 MB]      [Avail: 317.71 GB] Running on lane default
> [2026-06-09 15:17:04.13 GMT-4] [CPU: 79.2 MB]      [Avail: 317.70 GB] Resources allocated:
> [2026-06-09 15:17:04.13 GMT-4] [CPU: 79.2 MB]      [Avail: 317.70 GB]   Worker:  ganges.biology.columbia.edu
> [2026-06-09 15:17:04.14 GMT-4] [CPU: 79.2 MB]      [Avail: 317.70 GB]   CPU    :  [4, 5, 12, 13]
> [2026-06-09 15:17:04.14 GMT-4] [CPU: 79.2 MB]      [Avail: 317.70 GB]   GPU    :  [1]
> [2026-06-09 15:17:04.14 GMT-4] [CPU: 79.2 MB]      [Avail: 317.70 GB]   RAM    :  [2, 7, 8]
> [2026-06-09 15:17:04.15 GMT-4] [CPU: 79.2 MB]      [Avail: 317.70 GB]   SSD    :  True
> [2026-06-09 15:17:04.15 GMT-4] [CPU: 79.2 MB]      [Avail: 317.70 GB] -----
                                -
> [2026-06-09 15:17:04.15 GMT-4] [CPU: 79.2 MB]      [Avail: 317.70 GB] Importing job module for job type nonuniform_refine_new...
> [2026-06-09 15:17:18.35 GMT-4] [CPU: 273.6 MB]     [Avail: 317.01 GB] Job ready to run
> [2026-06-09 15:17:18.35 GMT-4] [CPU: 273.6 MB]     [Avail: 317.00 GB] *****
                                **
> [2026-06-09 15:17:18.55 GMT-4] [CPU: 287.9 MB]     [Avail: 316.78 GB] Using random seed of 57167689
> [2026-06-09 15:17:18.55 GMT-4] [CPU: 287.9 MB]     [Avail: 316.77 GB] Loading a ParticleStack with 5478 items...
> [2026-06-09 15:17:18.56 GMT-4] [CPU: 287.9 MB]     [Avail: 316.76 GB]

                                SSD cache ACTIVE at
                                /scr/instance_ganges.biology.columbia.edu:39001 (200 GB
                                reserve) (3 TB quota)

                                Cache usage                                Amount

                                Total / Usable                            3.58 TiB / 2.73 TiB
                                Used / Free                              2.73 TiB / 666.66 MiB
                                Hits / Misses                             45.80 GiB / 0.00 B
                                Acquired / Required                      45.80 GiB / 0.00 B

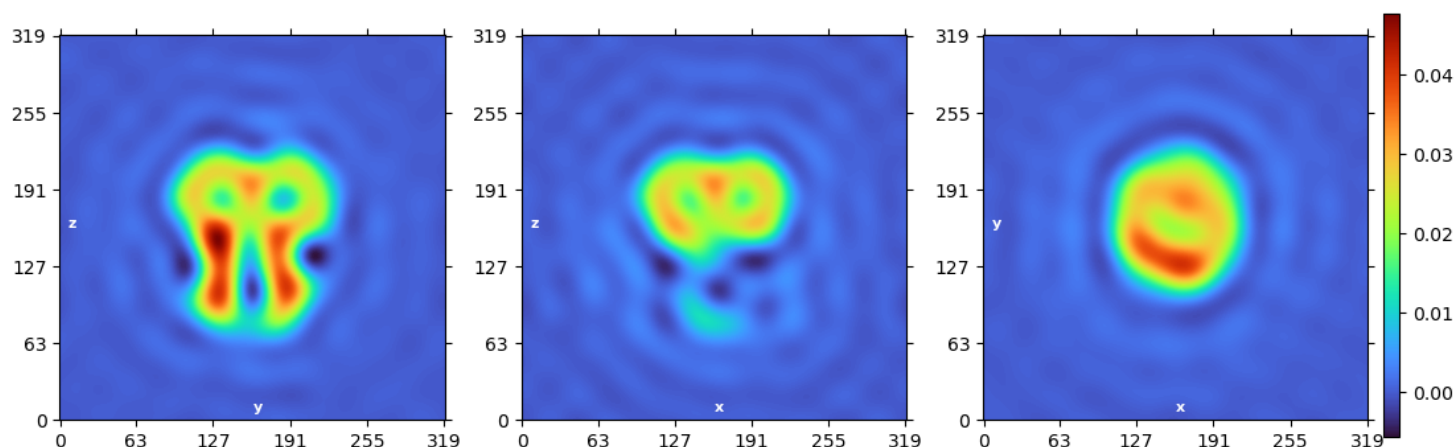
                                Progress:
                                [
                                3942/3942 (100%)
                                Elapsed: 0h 00m 04s
                                Active jobs: P25-J3397, P25-J3398
                                SSD cache complete for 3942 file(s)

> [2026-06-09 15:17:23.46 GMT-4] [CPU: 330.0 MB]     [Avail: 316.67 GB] Done.
> [2026-06-09 15:17:23.46 GMT-4] [CPU: 330.0 MB]     [Avail: 316.67 GB] Windowing particles
> [2026-06-09 15:17:23.47 GMT-4] [CPU: 331.2 MB]     [Avail: 316.67 GB] Done.
> [2026-06-09 15:17:23.47 GMT-4] [CPU: 331.2 MB]     [Avail: 316.67 GB] ===== Gold Standard Split =====
> [2026-06-09 15:17:23.48 GMT-4] [CPU: 331.2 MB]     [Avail: 316.67 GB] Particles have input alignments3D connected, so reusing
                                pre-existing split
> [2026-06-09 15:17:23.48 GMT-4] [CPU: 331.2 MB]     [Avail: 316.67 GB] Set A is greater than set B by 16 particles (0.292 percent
                                difference relative to the total dataset).
> [2026-06-09 15:17:23.50 GMT-4] [CPU: 331.7 MB]     [Avail: 316.66 GB] Split A has 2747 particles
> [2026-06-09 15:17:23.51 GMT-4] [CPU: 331.7 MB]     [Avail: 316.66 GB] Split B has 2731 particles
> [2026-06-09 15:17:23.51 GMT-4] [CPU: 331.7 MB]     [Avail: 316.66 GB] ===== Refinement =====
> [2026-06-09 15:17:23.51 GMT-4] [CPU: 331.7 MB]     [Avail: 316.66 GB] Input particles have box size 320
> [2026-06-09 15:17:23.52 GMT-4] [CPU: 331.7 MB]     [Avail: 316.66 GB] Input particles have pixel size 0.8300
> [2026-06-09 15:17:23.52 GMT-4] [CPU: 331.9 MB]     [Avail: 316.66 GB] Particles will be zeropadded/truncated to size 320 during
                                alignment
```

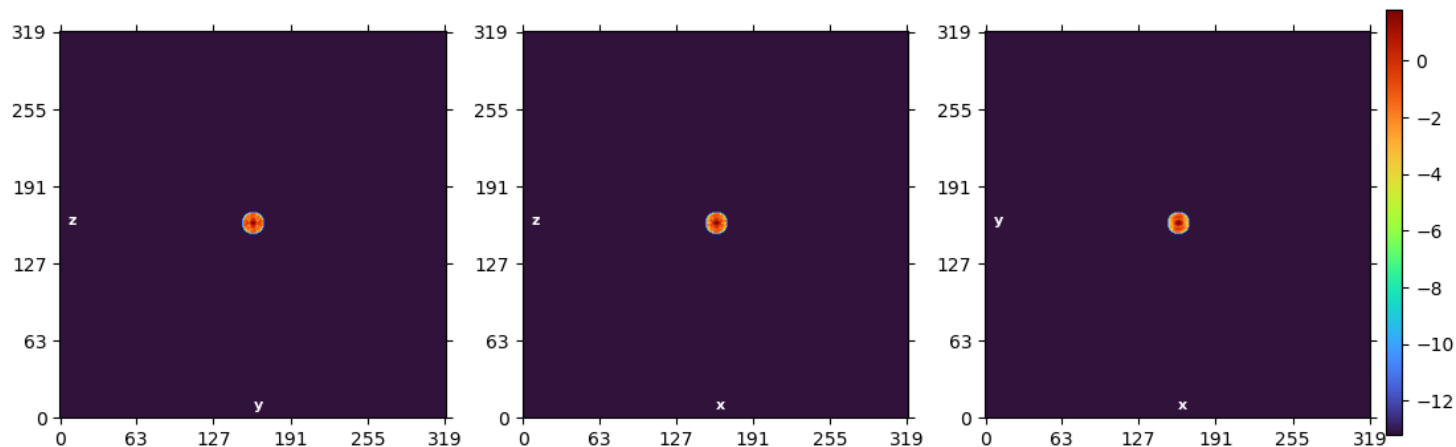
```

> [2026-06-09 15:17:23.52 GMT-4] [CPU: 331.9 MB] [Avail: 316.66 GB] Volume refinement will be done with effective box size 320
> [2026-06-09 15:17:23.52 GMT-4] [CPU: 331.9 MB] [Avail: 316.66 GB] Volume refinement will be done with pixel size 0.8300
> [2026-06-09 15:17:23.53 GMT-4] [CPU: 331.9 MB] [Avail: 316.66 GB] Particles will be zeropadded/truncated to size 320 during
backprojection
> [2026-06-09 15:17:23.53 GMT-4] [CPU: 331.9 MB] [Avail: 316.66 GB] Particles will be backprojected with box size 320
> [2026-06-09 15:17:23.53 GMT-4] [CPU: 331.9 MB] [Avail: 316.66 GB] Volume will be internally cropped and stored with box size
320
> [2026-06-09 15:17:23.54 GMT-4] [CPU: 331.9 MB] [Avail: 316.66 GB] Volume will be interpolated with box size 320 (zeropadding
factor 1.00)
> [2026-06-09 15:17:23.54 GMT-4] [CPU: 331.9 MB] [Avail: 316.66 GB] DC components of images will be ignored and volume will be
floated at each iteration.
> [2026-06-09 15:17:23.54 GMT-4] [CPU: 331.9 MB] [Avail: 316.66 GB] Spherical windowing of maps is enabled
> [2026-06-09 15:17:23.55 GMT-4] [CPU: 331.9 MB] [Avail: 316.66 GB] Refining with C1 symmetry enforced
> [2026-06-09 15:17:23.55 GMT-4] [CPU: 331.9 MB] [Avail: 316.66 GB] Refining with pose and shift marginalization during
backprojection enabled.
> [2026-06-09 15:17:23.55 GMT-4] [CPU: 331.9 MB] [Avail: 316.66 GB] Resetting input per-particle scale factors to 1.0
> [2026-06-09 15:17:23.55 GMT-4] [CPU: 331.9 MB] [Avail: 316.66 GB] Starting at initial resolution 30.000A (radwn 8.853).
> [2026-06-09 15:17:23.56 GMT-4] [CPU: 331.9 MB] [Avail: 316.66 GB] ===== Non-Uniform Refinement =====
> [2026-06-09 15:17:23.56 GMT-4] [CPU: 331.9 MB] [Avail: 316.66 GB] Non-Uniform Refinement is enabled.
> [2026-06-09 15:17:23.56 GMT-4] [CPU: 331.9 MB] [Avail: 316.66 GB] Using AWF of 3.00.
> [2026-06-09 15:17:23.57 GMT-4] [CPU: 331.9 MB] [Avail: 316.66 GB] Using butterworth filter with order 8.
> [2026-06-09 15:17:23.57 GMT-4] [CPU: 332.1 MB] [Avail: 316.66 GB] ===== Masking =====
> [2026-06-09 15:17:25.83 GMT-4] [CPU: 864.6 MB] [Avail: 316.03 GB] No mask input was connected, so dynamic masking will be
enabled.
> [2026-06-09 15:17:25.83 GMT-4] [CPU: 864.6 MB] [Avail: 316.03 GB] Dynamic mask threshold: 0.2000
> [2026-06-09 15:17:25.83 GMT-4] [CPU: 864.6 MB] [Avail: 316.03 GB] Dynamic mask near (A): 6.00
> [2026-06-09 15:17:25.84 GMT-4] [CPU: 864.6 MB] [Avail: 316.03 GB] Dynamic mask far (A): 14.00
> [2026-06-09 15:17:25.84 GMT-4] [CPU: 864.6 MB] [Avail: 316.02 GB] ===== Initial Model =====
> [2026-06-09 15:17:25.84 GMT-4] [CPU: 864.6 MB] [Avail: 316.03 GB] Resampling initial model to specified volume representation
size and pixel-size...
> [2026-06-09 15:17:29.03 GMT-4] [CPU: 1.25 GB] [Avail: 317.37 GB] Estimating scale of initial reference.
> [2026-06-09 15:17:33.95 GMT-4] [CPU: 77.4 MB] [Avail: 314.30 GB] WARNING: io_uring support disabled (not supported by kernel),
I/O performance may degrade
> [2026-06-09 15:17:35.67 GMT-4] [CPU: 1.45 GB] [Avail: 316.68 GB] Rescaling initial reference by a factor of 1.435
> [2026-06-09 15:17:37.35 GMT-4] [CPU: 1.45 GB] [Avail: 316.48 GB] Estimating scale of initial reference.
> [2026-06-09 15:17:40.43 GMT-4] [CPU: 1.41 GB] [Avail: 315.26 GB] Rescaling initial reference by a factor of 0.992
> [2026-06-09 15:17:41.89 GMT-4] [CPU: 1.41 GB] [Avail: 315.54 GB] Estimating scale of initial reference.
> [2026-06-09 15:17:44.66 GMT-4] [CPU: 1.41 GB] [Avail: 314.17 GB] Rescaling initial reference by a factor of 1.007
> [2026-06-09 15:17:47.58 GMT-4] Initial Real Space Slices

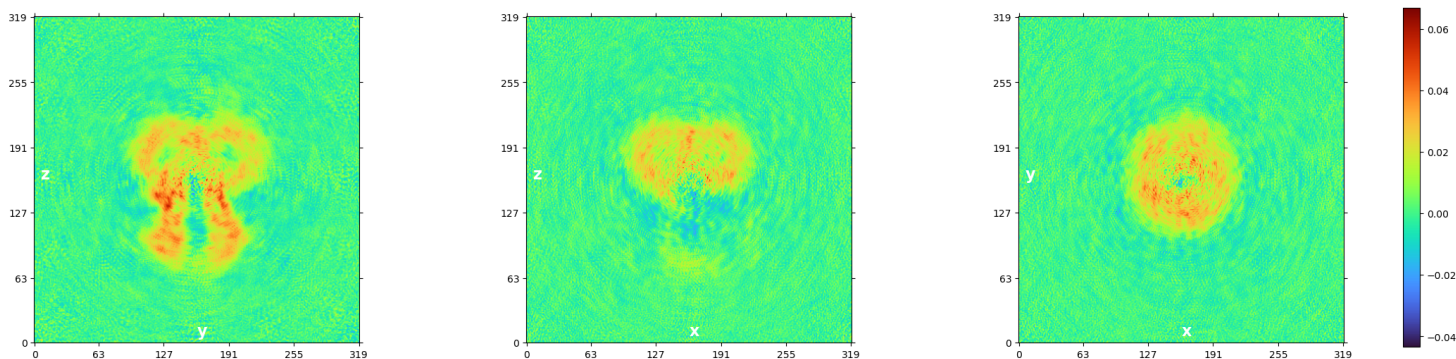
```



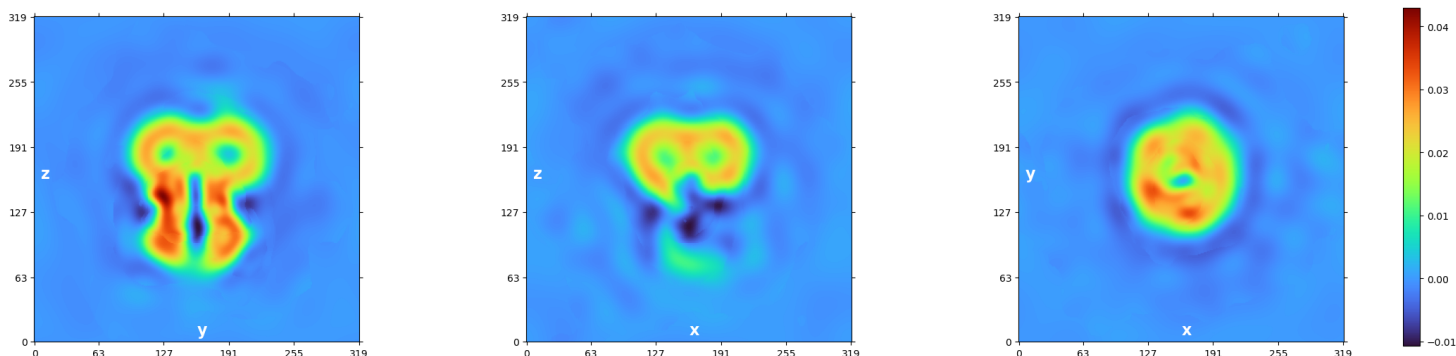
> [2026-06-09 15:17:49.13 GMT-4] Initial Fourier Space Slices



```
> [2026-06-09 15:17:49.14 GMT-4] [CPU: 1.46 GB] [Avail: 312.68 GB] ===== Starting Refinement Iterations =====
> [2026-06-09 15:17:49.14 GMT-4] [CPU: 1.46 GB] [Avail: 312.68 GB] ----- Start Iteration 0
> [2026-06-09 15:17:49.14 GMT-4] [CPU: 1.46 GB] [Avail: 312.68 GB] Using Max Alignment Radius 8.853 (30.000A)
> [2026-06-09 15:17:49.15 GMT-4] [CPU: 1.46 GB] [Avail: 312.68 GB] Auto batchsize: 2747 in each split
> [2026-06-09 15:17:52.67 GMT-4] [CPU: 1.97 GB] [Avail: 313.46 GB] -- THR 1 BATCH 500 NUM 866 TOTAL 2.1425867 ELAPSED 21.377187
--
> [2026-06-09 15:18:16.01 GMT-4] [CPU: 3.15 GB] [Avail: 306.54 GB] Processed 5478.000 images in 23.342s.
> [2026-06-09 15:18:19.84 GMT-4] [CPU: 3.48 GB] [Avail: 309.22 GB] Computing FSCs...
> [2026-06-09 15:18:19.84 GMT-4] [CPU: 3.48 GB] [Avail: 309.21 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:18:19.85 GMT-4] [CPU: 3.48 GB] [Avail: 309.20 GB] Computing FFTs on GPU.
> [2026-06-09 15:18:22.27 GMT-4] [CPU: 3.73 GB] [Avail: 307.35 GB] Done in 2.425s
> [2026-06-09 15:18:22.27 GMT-4] [CPU: 3.73 GB] [Avail: 307.35 GB] Computing cFSCs...
> [2026-06-09 15:18:24.16 GMT-4] [CPU: 3.73 GB] [Avail: 307.08 GB] Done in 1.879s
> [2026-06-09 15:18:24.16 GMT-4] [CPU: 3.73 GB] [Avail: 307.07 GB] Using Filter Radius 25.652 (10.354A) | Previous: 8.853
(30.000A)
> [2026-06-09 15:18:29.95 GMT-4] [CPU: 4.67 GB] [Avail: 306.35 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:18:29.95 GMT-4] [CPU: 4.67 GB] [Avail: 306.38 GB] Running local cross validation for A ...
> [2026-06-09 15:18:58.40 GMT-4] [CPU: 7.50 GB] [Avail: 302.89 GB] Local cross validation A done in 28.449s
> [2026-06-09 15:19:00.19 GMT-4] FSC Filtered Side A
```

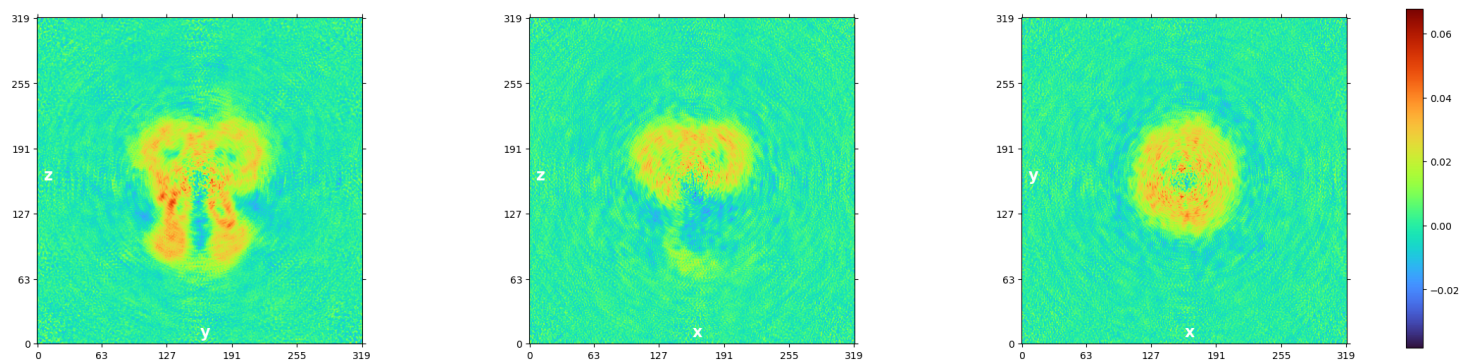


> [2026-06-09 15:19:01.73 GMT-4] CV Filtered Side A

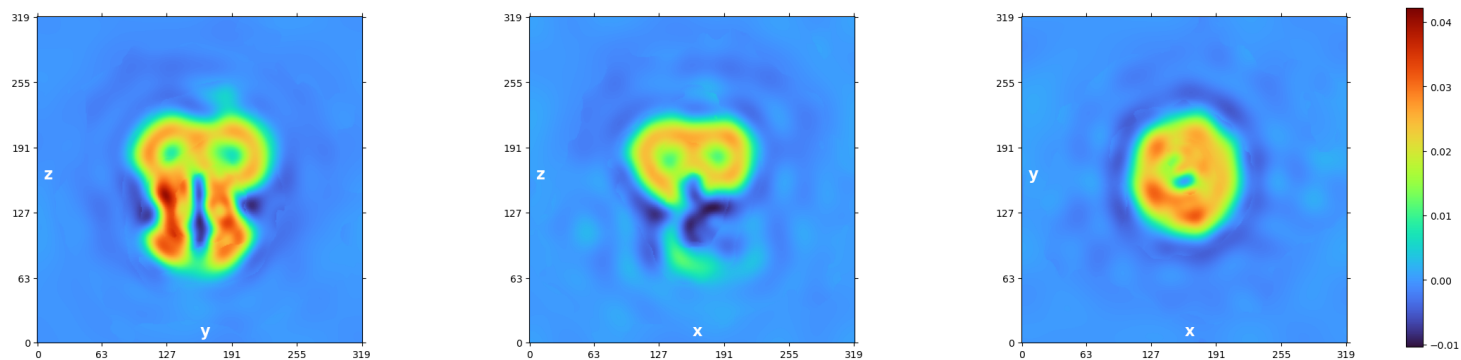


```
> [2026-06-09 15:19:01.73 GMT-4] [CPU: 7.51 GB] [Avail: 300.60 GB] Running local cross validation for B ...
> [2026-06-09 15:19:27.99 GMT-4] [CPU: 7.79 GB] [Avail: 299.28 GB] Local cross validation B done in 26.252s
```


> [2026-06-09 15:19:29.48 GMT-4] FSC Filtered Side B

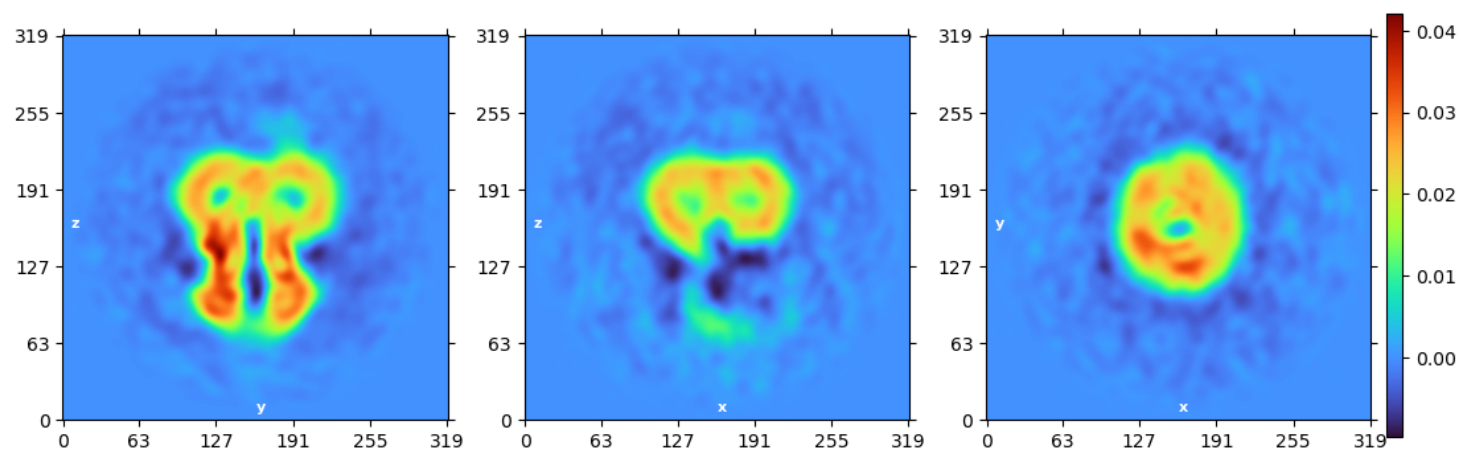


> [2026-06-09 15:19:30.83 GMT-4] CV Filtered Side B

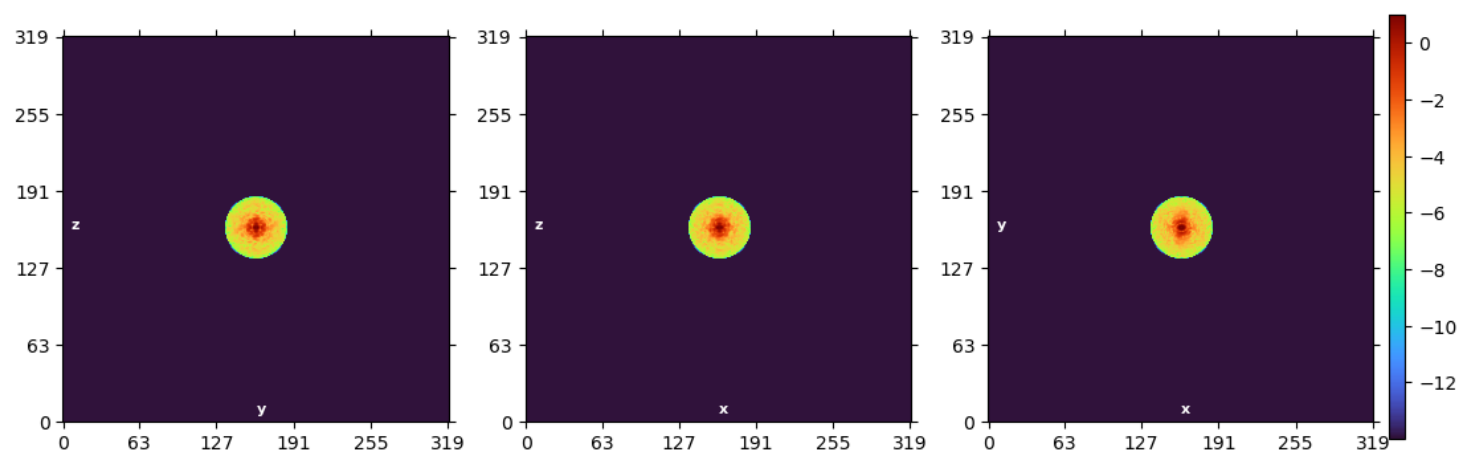


> [2026-06-09 15:19:36.56 GMT-4] [CPU: 8.02 GB] [Avail: 298.61 GB] Plotting..

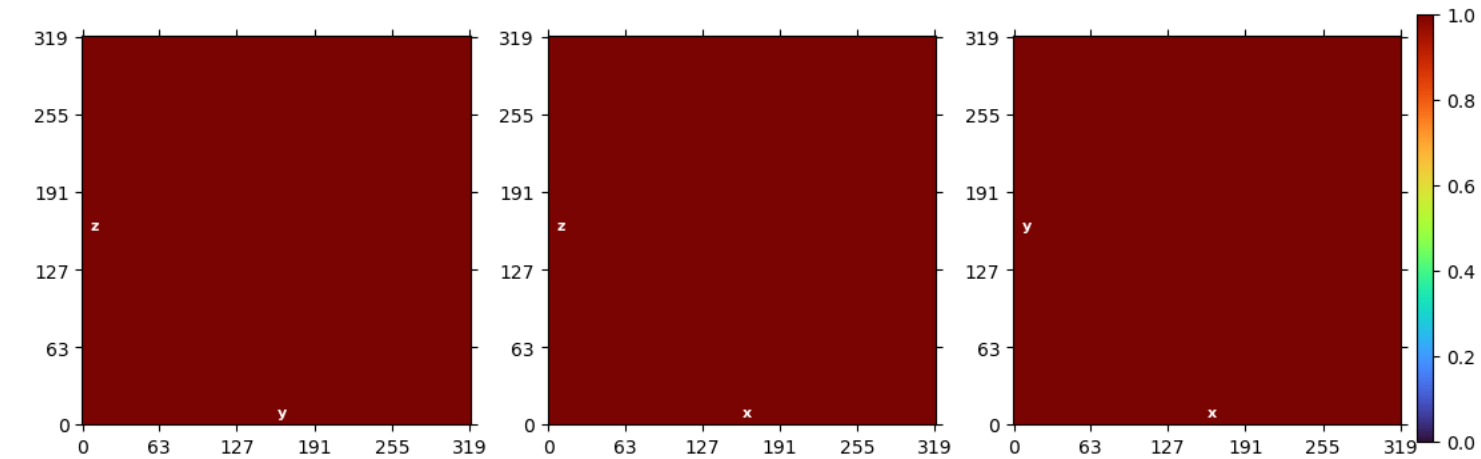
> [2026-06-09 15:19:43.40 GMT-4] Real Space Slices Iteration 000



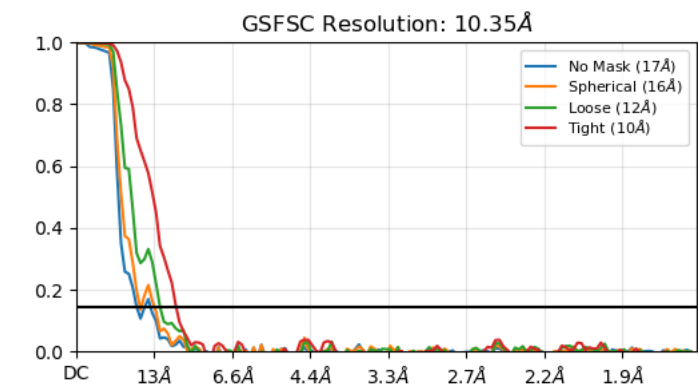
> [2026-06-09 15:19:44.89 GMT-4] Fourier Space Slices Iteration 000



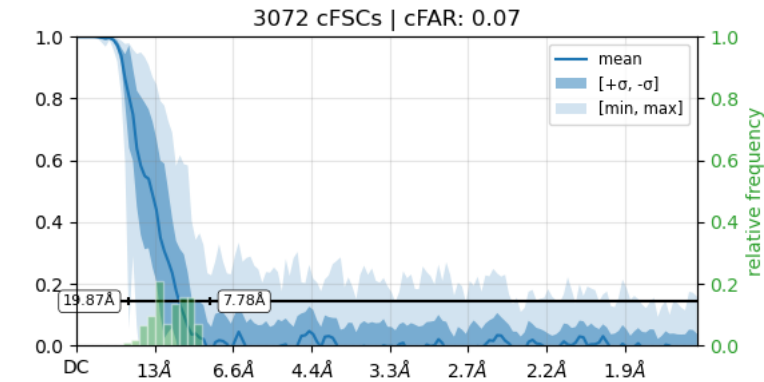
> [2026-06-09 15:19:46.15 GMT-4] Real Space Mask Slices Iteration 000



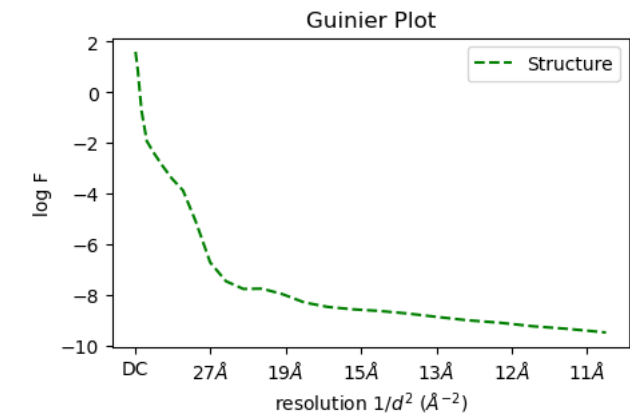
> [2026-06-09 15:19:46.87 GMT-4] FSC Iteration 000



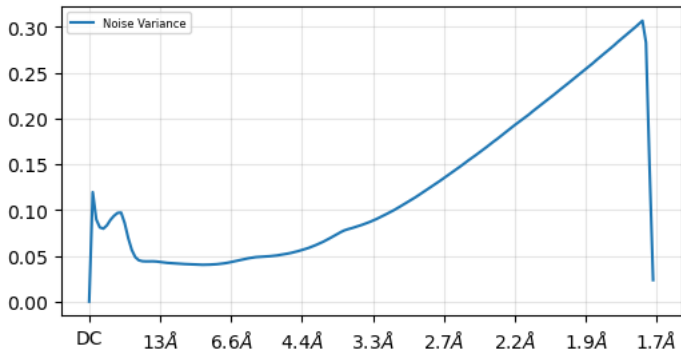
> [2026-06-09 15:19:48.39 GMT-4] cFSCs (Half-angle: 20°) Iteration 000, with tight mask



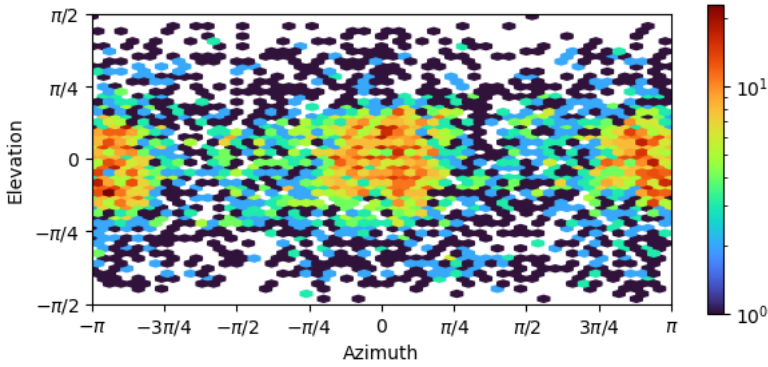
> [2026-06-09 15:19:53.88 GMT-4] Guinier Plot Iteration 000



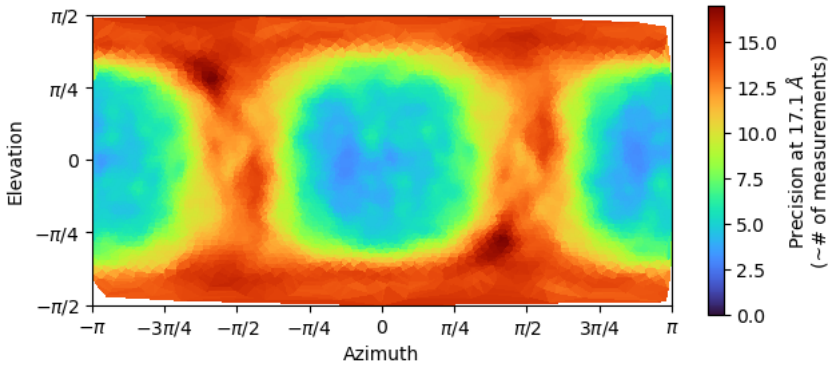
> [2026-06-09 15:19:54.31 GMT-4] Noise Model Iteration 000



> [2026-06-09 15:19:55.13 GMT-4] Viewing Direction Distribution Iteration 000

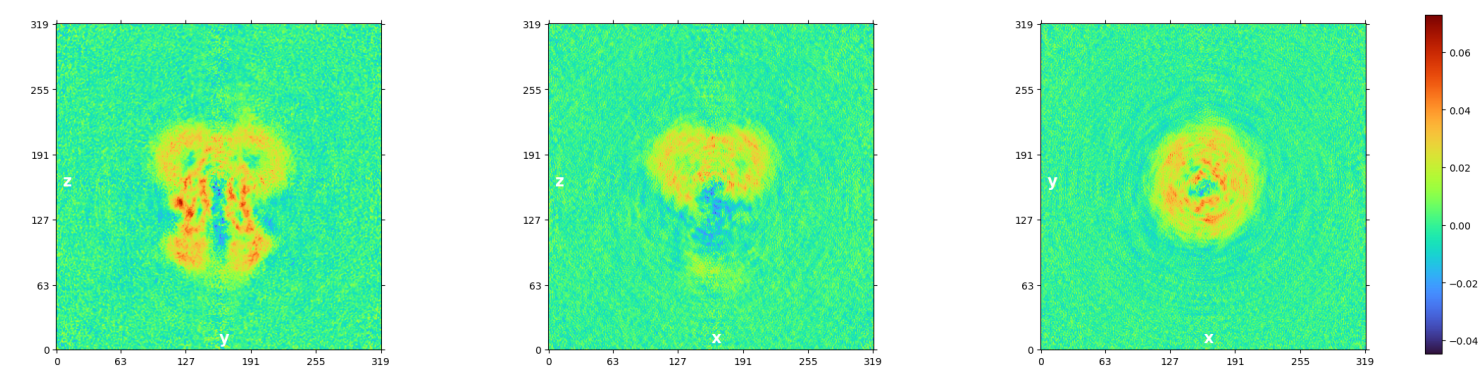


> [2026-06-09 15:19:57.09 GMT-4] Posterior Precision Directional Distribution Iteration 000

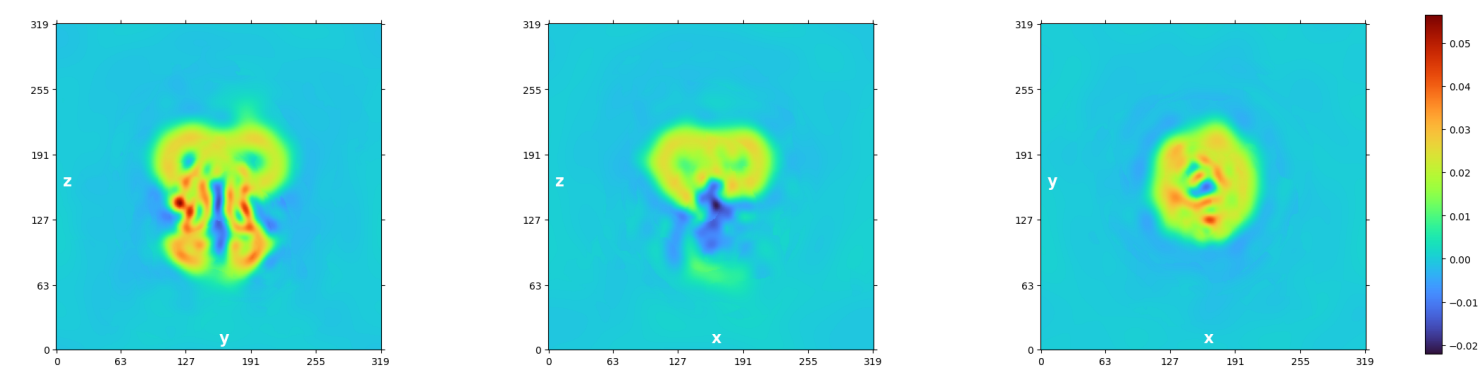


```
> [2026-06-09 15:19:57.09 GMT-4] [CPU: 8.45 GB] [Avail: 296.96 GB] Done in 20.528s.
> [2026-06-09 15:19:57.10 GMT-4] [CPU: 8.45 GB] [Avail: 296.96 GB] Outputting files..
> [2026-06-09 15:20:11.11 GMT-4] [CPU: 8.88 GB] [Avail: 299.74 GB] Done in 14.015s.
> [2026-06-09 15:20:11.12 GMT-4] [CPU: 8.88 GB] [Avail: 299.74 GB] Done iteration 0 in 141.974s. Total time so far 172.731s
> [2026-06-09 15:20:11.65 GMT-4] [CPU: 8.88 GB] [Avail: 299.94 GB] ----- Start Iteration 1
> [2026-06-09 15:20:11.66 GMT-4] [CPU: 8.88 GB] [Avail: 299.94 GB] Using Max Alignment Radius 25.652 (10.354A)
> [2026-06-09 15:20:11.66 GMT-4] [CPU: 8.88 GB] [Avail: 299.93 GB] Auto batchsize: 2747 in each split
> [2026-06-09 15:20:11.66 GMT-4] [CPU: 8.88 GB] [Avail: 299.93 GB] Using dynamic mask.
> [2026-06-09 15:20:39.54 GMT-4] [CPU: 9.38 GB] [Avail: 297.88 GB] -- THR 0 BATCH 500 NUM 866 TOTAL 2.3630802 ELAPSED 18.863724
--
> [2026-06-09 15:21:00.69 GMT-4] [CPU: 10.24 GB] [Avail: 296.93 GB] Processed 5478.000 images in 21.150s.
> [2026-06-09 15:21:04.10 GMT-4] [CPU: 10.24 GB] [Avail: 296.59 GB] Computing FSCs...
> [2026-06-09 15:21:04.11 GMT-4] [CPU: 10.24 GB] [Avail: 296.59 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:21:04.11 GMT-4] [CPU: 10.24 GB] [Avail: 296.59 GB] Computing FFTs on GPU.
> [2026-06-09 15:21:06.10 GMT-4] [CPU: 10.24 GB] [Avail: 295.74 GB] Done in 1.993s
> [2026-06-09 15:21:06.11 GMT-4] [CPU: 10.24 GB] [Avail: 295.73 GB] Computing cFSCs...
> [2026-06-09 15:21:08.36 GMT-4] [CPU: 10.24 GB] [Avail: 296.79 GB] Done in 2.250s
> [2026-06-09 15:21:08.37 GMT-4] [CPU: 10.24 GB] [Avail: 296.82 GB] Using Filter Radius 31.953 (8.312A) | Previous: 25.652
(10.354A)
> [2026-06-09 15:21:14.80 GMT-4] [CPU: 9.48 GB] [Avail: 295.54 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:21:14.80 GMT-4] [CPU: 9.48 GB] [Avail: 295.54 GB] Running local cross validation for A ...
> [2026-06-09 15:21:37.10 GMT-4] [CPU: 9.74 GB] [Avail: 291.08 GB] Local cross validation A done in 22.296s
```


> [2026-06-09 15:21:38.59 GMT-4] FSC Filtered Side A



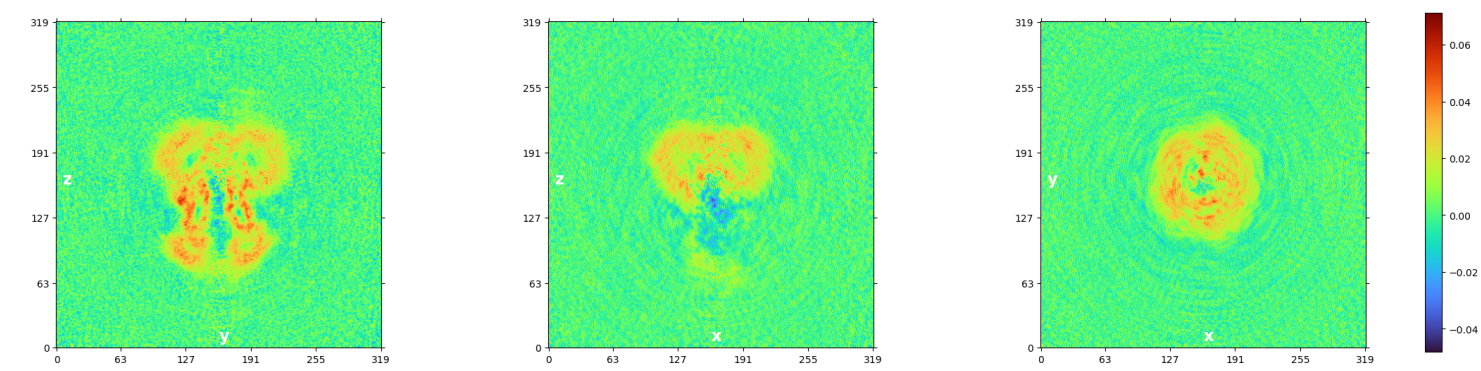
> [2026-06-09 15:21:39.62 GMT-4] CV Filtered Side A



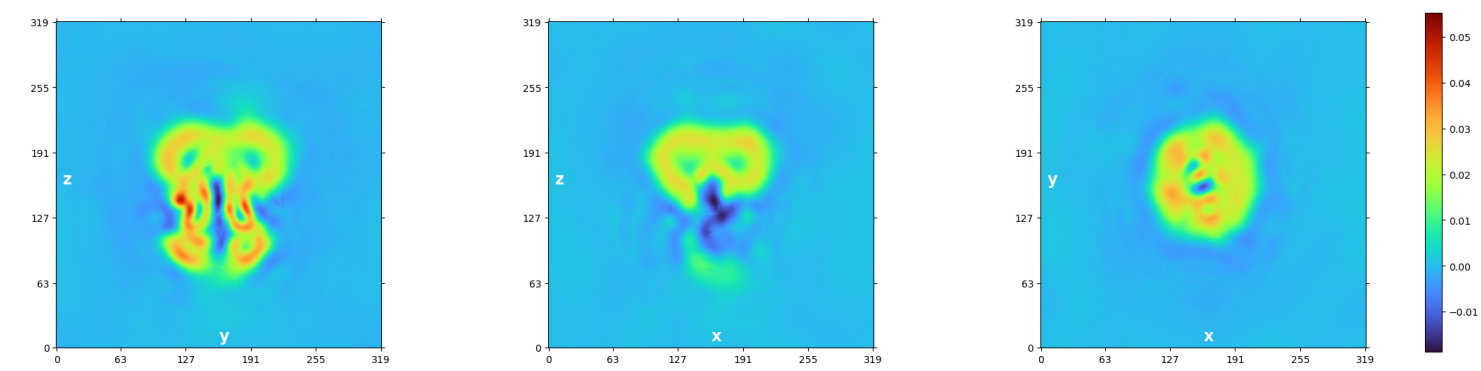
> [2026-06-09 15:21:39.63 GMT-4] [CPU: 9.74 GB] [Avail: 291.92 GB] Running local cross validation for B ...

> [2026-06-09 15:22:01.57 GMT-4] [CPU: 9.99 GB] [Avail: 294.47 GB] Local cross validation B done in 21.945s

> [2026-06-09 15:22:02.94 GMT-4] FSC Filtered Side B



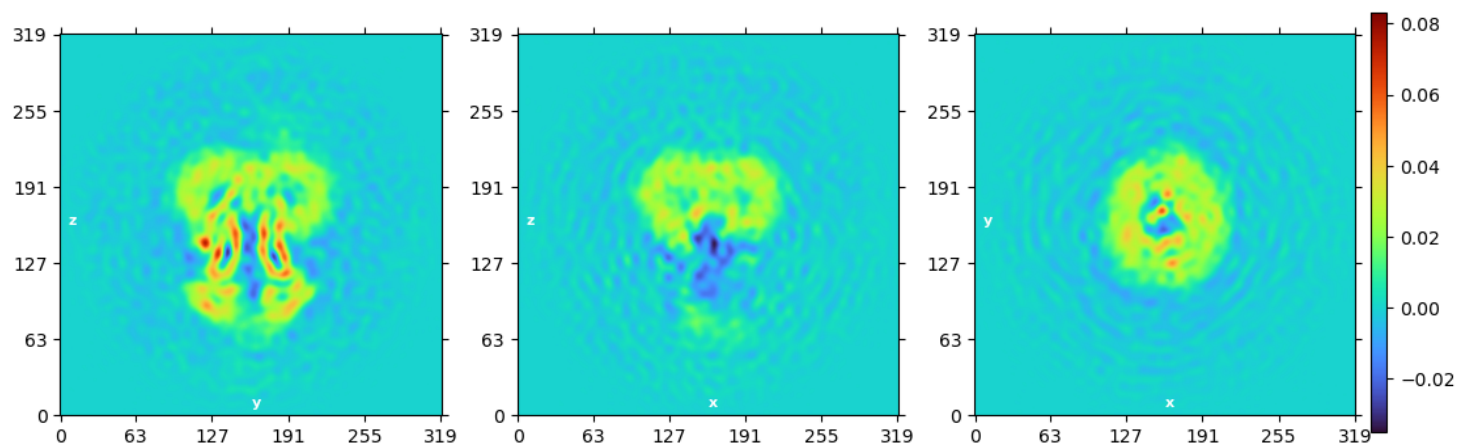
> [2026-06-09 15:22:03.98 GMT-4] CV Filtered Side B



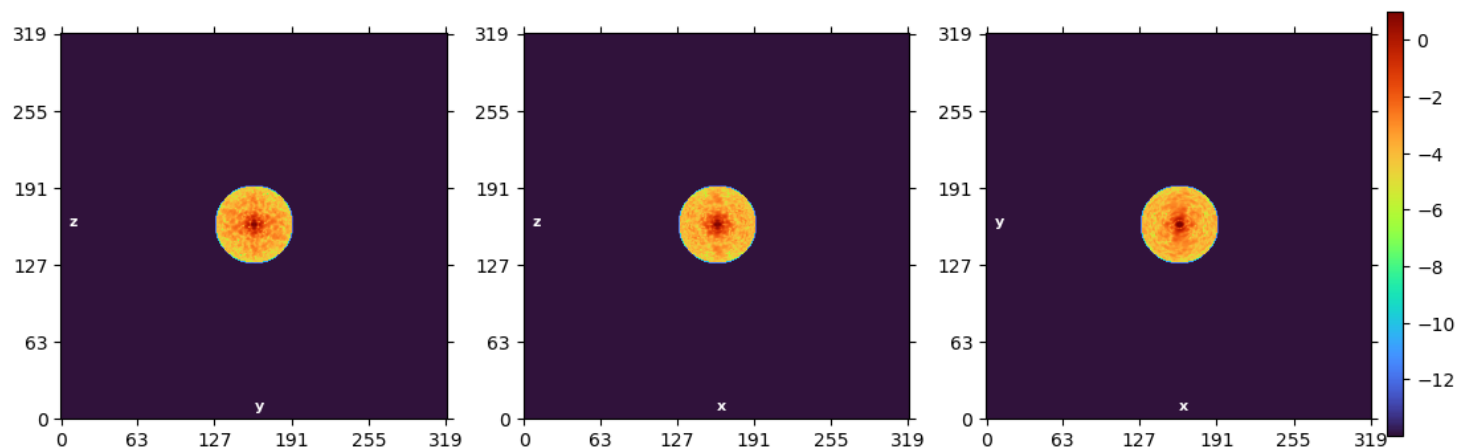
> [2026-06-09 15:22:15.24 GMT-4] [CPU: 10.20 GB] [Avail: 291.27 GB] Estimated Bfactor: -614.3

> [2026-06-09 15:22:15.25 GMT-4] [CPU: 10.20 GB] [Avail: 291.27 GB] Plotting..

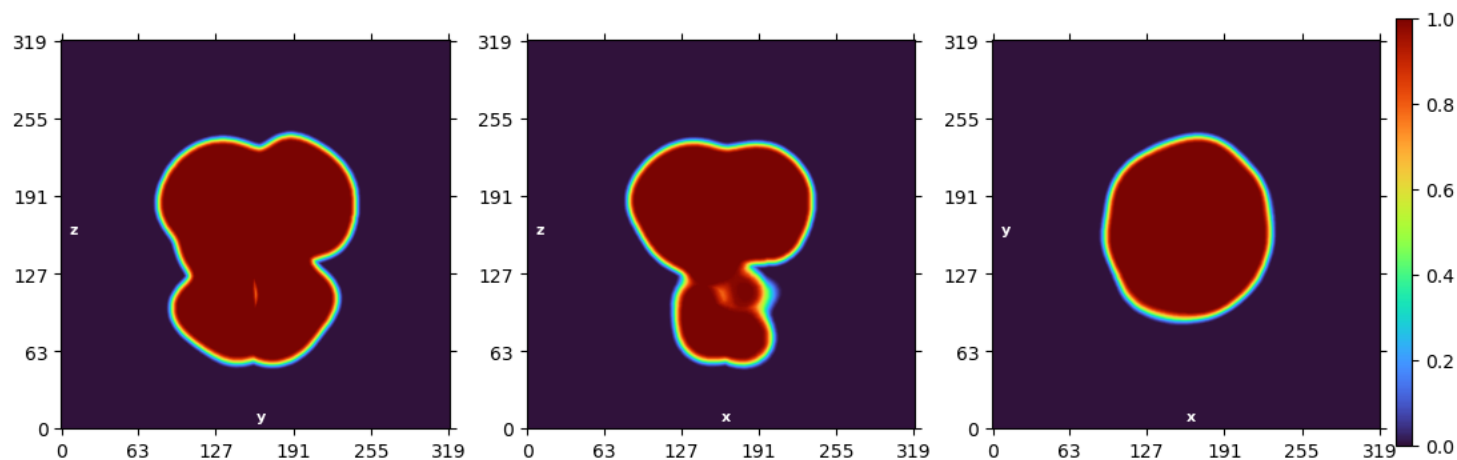
> [2026-06-09 15:22:21.95 GMT-4] Real Space Slices Iteration 001



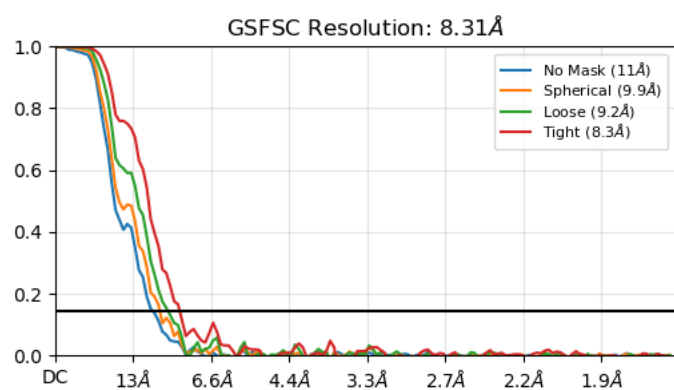
> [2026-06-09 15:22:23.20 GMT-4] Fourier Space Slices Iteration 001



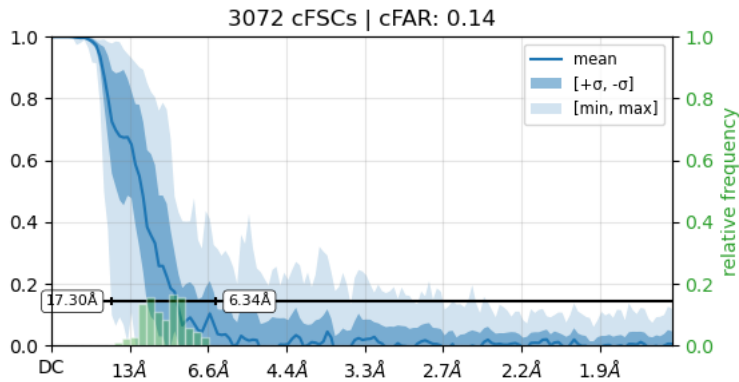
> [2026-06-09 15:22:25.14 GMT-4] Real Space Mask Slices Iteration 001



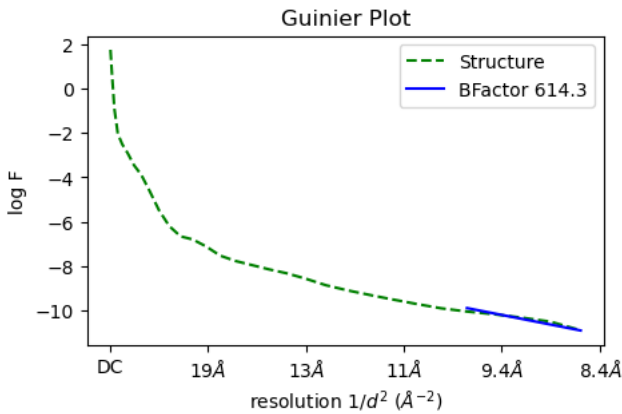
> [2026-06-09 15:22:25.59 GMT-4] FSC Iteration 001



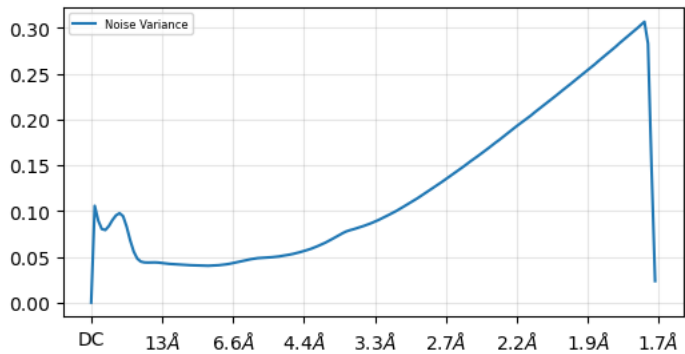
> [2026-06-09 15:22:26.26 GMT-4] cFSCs (Half-angle: 20°) Iteration 001, with tight mask



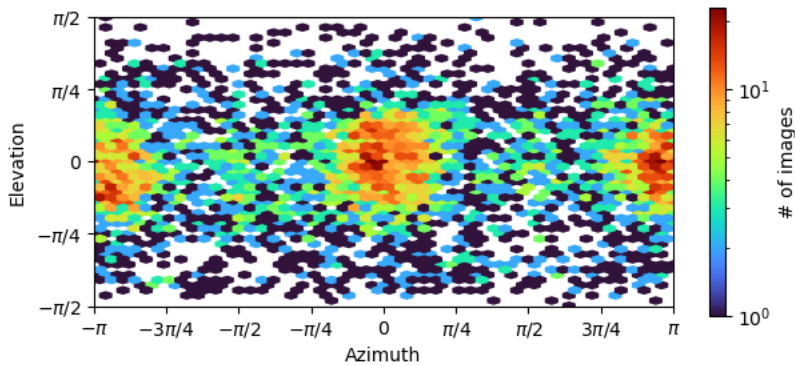
> [2026-06-09 15:22:32.15 GMT-4] Guinier Plot Iteration 001



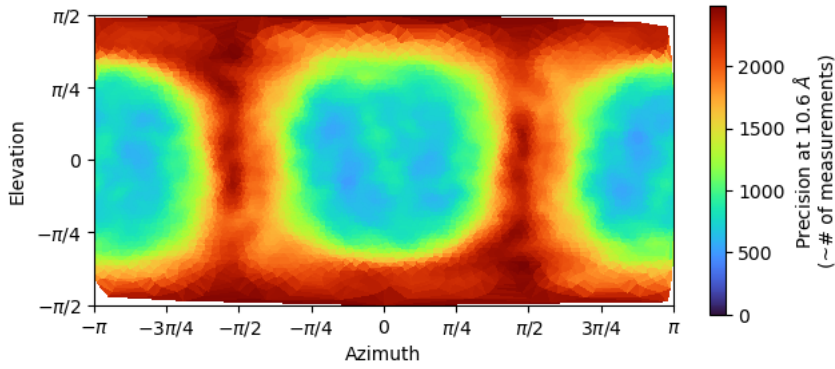
> [2026-06-09 15:22:32.61 GMT-4] Noise Model Iteration 001



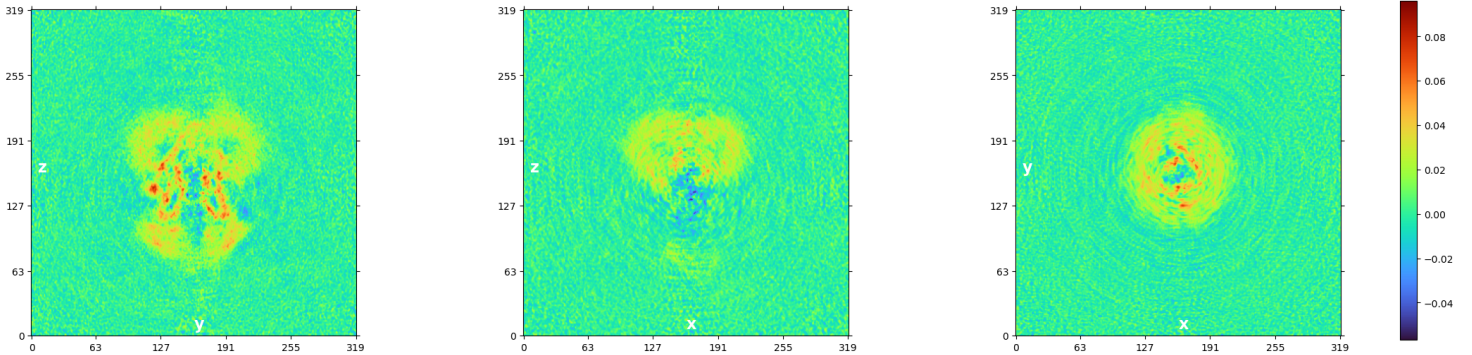
> [2026-06-09 15:22:33.59 GMT-4] Viewing Direction Distribution Iteration 001



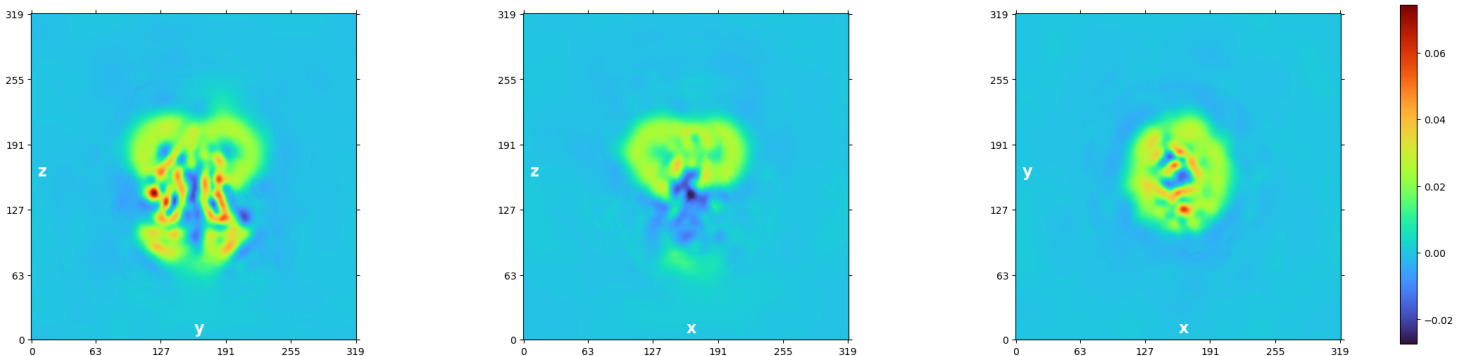
> [2026-06-09 15:22:35.57 GMT-4] Posterior Precision Directional Distribution Iteration 001



```
> [2026-06-09 15:22:35.58 GMT-4] [CPU: 10.24 GB] [Avail: 293.12 GB] Done in 20.324s.
> [2026-06-09 15:22:35.58 GMT-4] [CPU: 10.24 GB] [Avail: 293.12 GB] Outputting files..
> [2026-06-09 15:22:52.17 GMT-4] [CPU: 10.15 GB] [Avail: 294.00 GB] Done in 16.584s.
> [2026-06-09 15:22:52.17 GMT-4] [CPU: 10.15 GB] [Avail: 294.00 GB] Done iteration 1 in 160.513s. Total time so far 333.782s
> [2026-06-09 15:22:53.47 GMT-4] [CPU: 10.15 GB] [Avail: 293.82 GB] ----- Start Iteration 2
> [2026-06-09 15:22:53.48 GMT-4] [CPU: 10.15 GB] [Avail: 293.82 GB] Using Max Alignment Radius 31.953 (8.312A)
> [2026-06-09 15:22:53.48 GMT-4] [CPU: 10.15 GB] [Avail: 293.82 GB] Auto batchsize: 2747 in each split
> [2026-06-09 15:22:53.49 GMT-4] [CPU: 10.15 GB] [Avail: 293.82 GB] Using dynamic mask.
> [2026-06-09 15:23:25.14 GMT-4] [CPU: 10.14 GB] [Avail: 294.34 GB] -- THR 0 BATCH 500 NUM 866 TOTAL 2.9490604 ELAPSED 19.257007
--
> [2026-06-09 15:23:46.99 GMT-4] [CPU: 11.00 GB] [Avail: 295.04 GB] Processed 5478.000 images in 21.852s.
> [2026-06-09 15:23:50.52 GMT-4] [CPU: 11.00 GB] [Avail: 294.65 GB] Computing FSCs...
> [2026-06-09 15:23:50.52 GMT-4] [CPU: 11.00 GB] [Avail: 294.65 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:23:50.53 GMT-4] [CPU: 11.00 GB] [Avail: 294.65 GB] Computing FFTs on GPU.
> [2026-06-09 15:23:52.50 GMT-4] [CPU: 11.00 GB] [Avail: 293.80 GB] Done in 1.975s
> [2026-06-09 15:23:52.50 GMT-4] [CPU: 11.00 GB] [Avail: 293.79 GB] Computing cFSCs...
> [2026-06-09 15:23:54.55 GMT-4] [CPU: 11.00 GB] [Avail: 293.32 GB] Done in 2.046s
> [2026-06-09 15:23:54.56 GMT-4] [CPU: 11.00 GB] [Avail: 293.32 GB] Using Filter Radius 34.069 (7.796A) | Previous: 31.953
(8.312A)
> [2026-06-09 15:24:01.33 GMT-4] [CPU: 10.25 GB] [Avail: 295.00 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:24:01.34 GMT-4] [CPU: 10.25 GB] [Avail: 295.00 GB] Running local cross validation for A ...
> [2026-06-09 15:24:23.66 GMT-4] [CPU: 10.50 GB] [Avail: 291.85 GB] Local cross validation A done in 22.316s
> [2026-06-09 15:24:25.12 GMT-4] FSC Filtered Side A
```



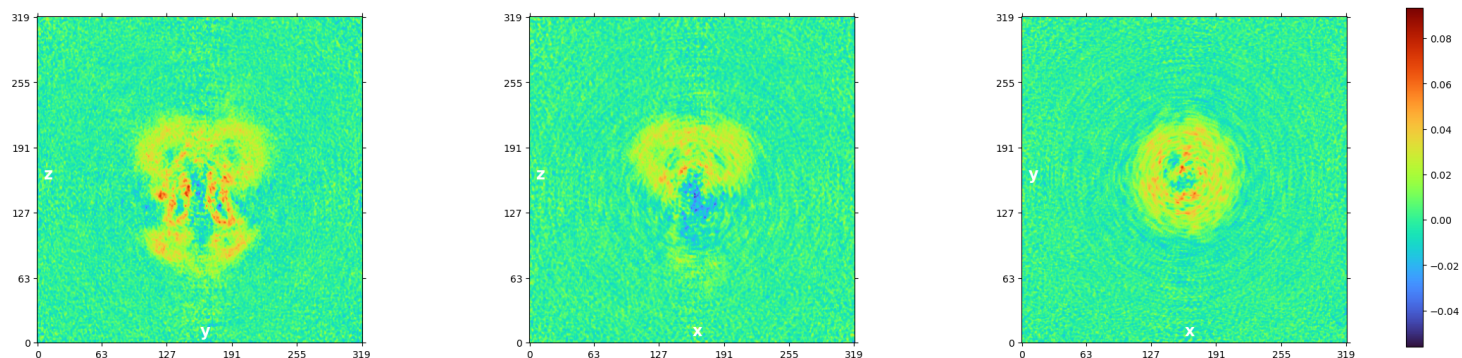
> [2026-06-09 15:24:26.50 GMT-4] CV Filtered Side A



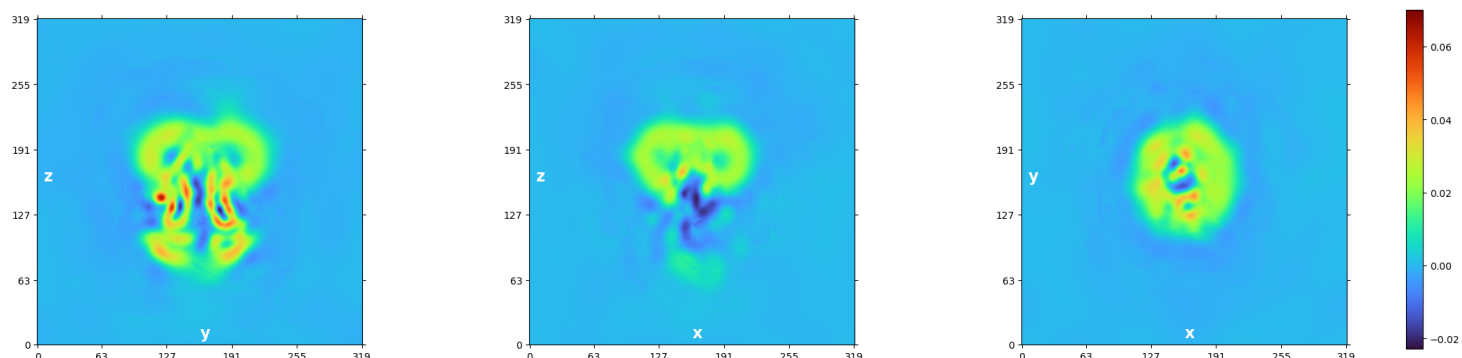
> [2026-06-09 15:24:26.50 GMT-4] [CPU: 10.50 GB] [Avail: 291.02 GB] Running local cross validation for B ...

> [2026-06-09 15:24:48.55 GMT-4] [CPU: 10.75 GB] [Avail: 293.47 GB] Local cross validation B done in 22.048s

> [2026-06-09 15:24:50.08 GMT-4] FSC Filtered Side B



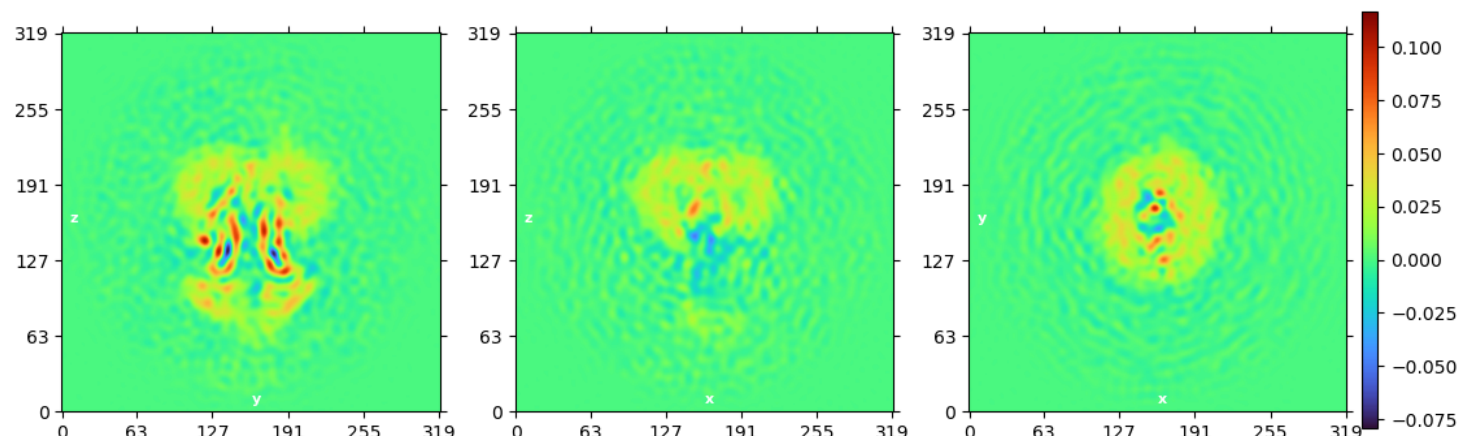
> [2026-06-09 15:24:51.24 GMT-4] CV Filtered Side B



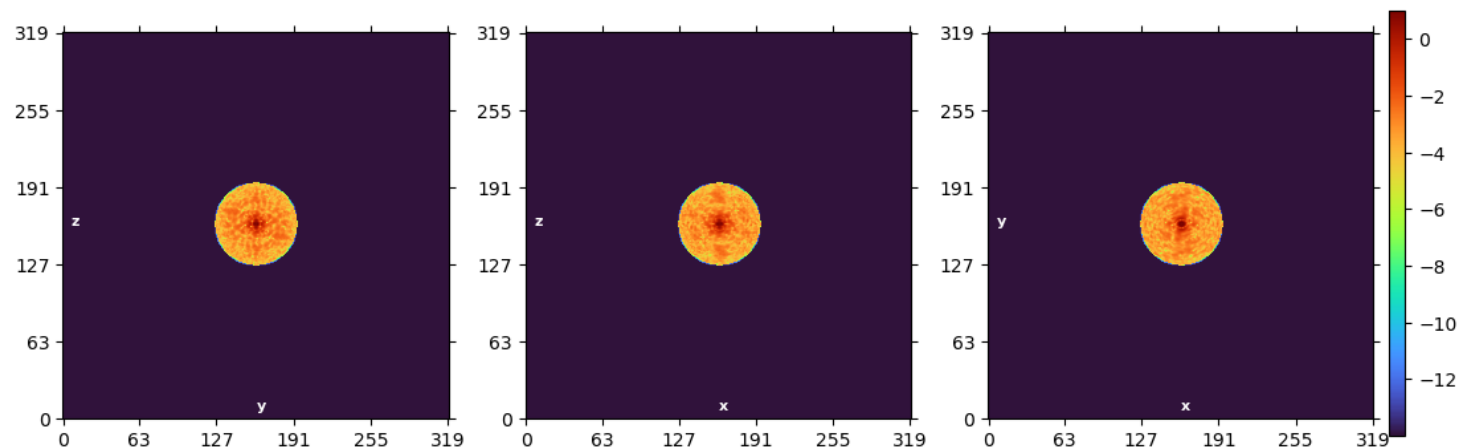
> [2026-06-09 15:25:02.20 GMT-4] [CPU: 10.72 GB] [Avail: 292.34 GB] Estimated Bfactor: -582.5

> [2026-06-09 15:25:02.21 GMT-4] [CPU: 10.72 GB] [Avail: 292.33 GB] Plotting..

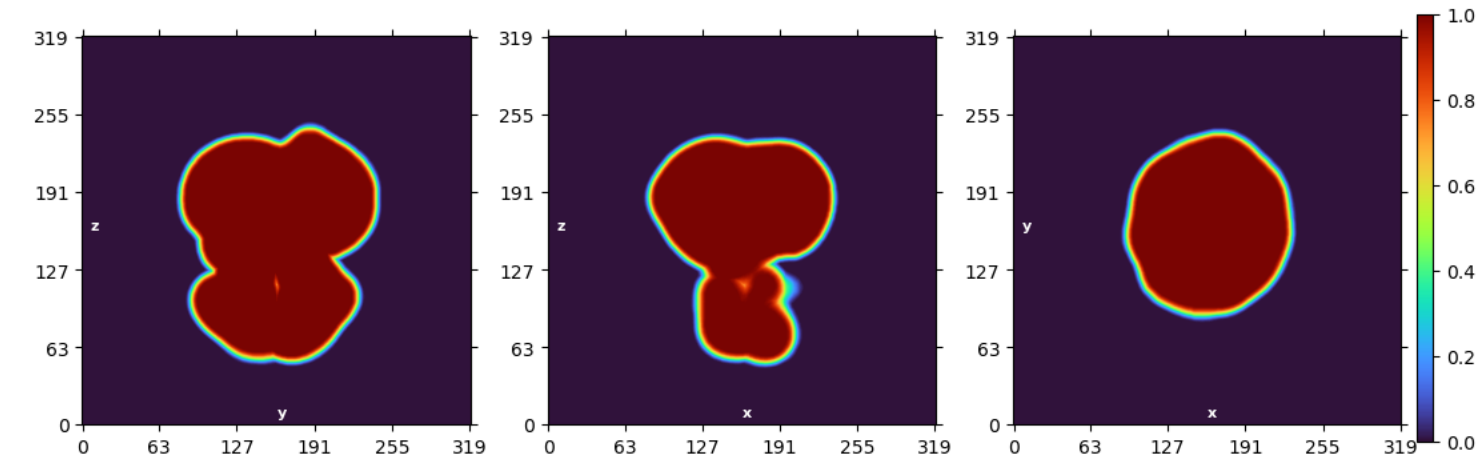
> [2026-06-09 15:25:09.15 GMT-4] Real Space Slices Iteration 002



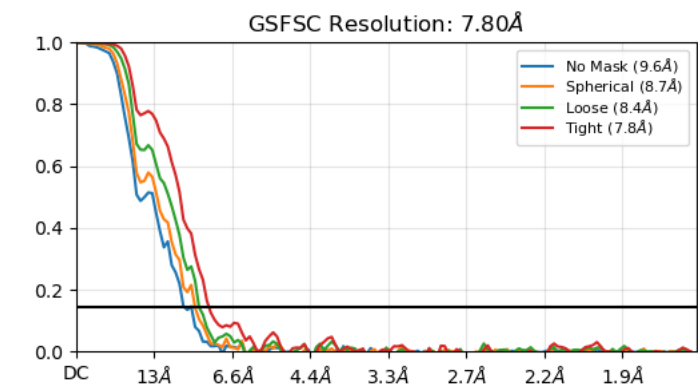
> [2026-06-09 15:25:10.64 GMT-4] Fourier Space Slices Iteration 002



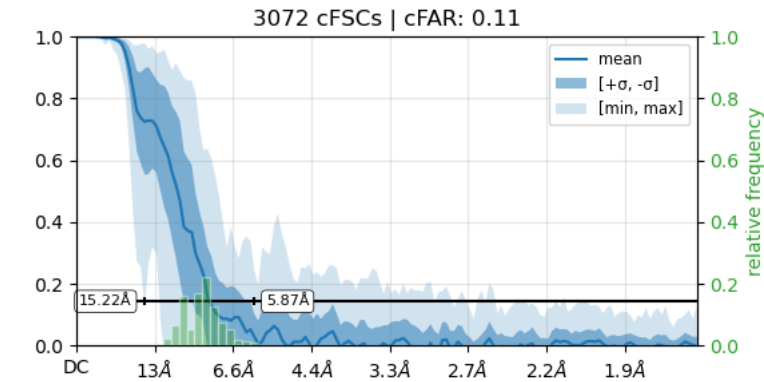
> [2026-06-09 15:25:12.92 GMT-4] Real Space Mask Slices Iteration 002



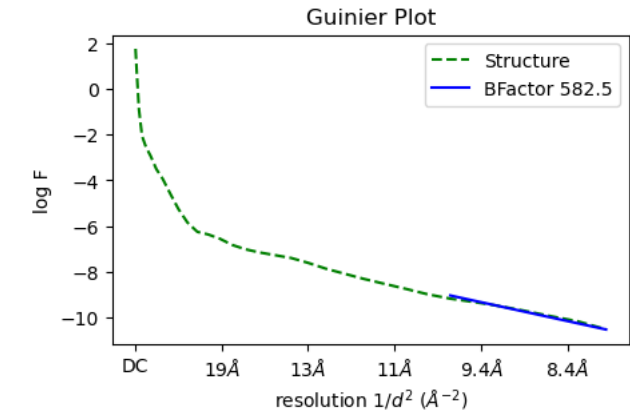
> [2026-06-09 15:25:13.55 GMT-4] FSC Iteration 002



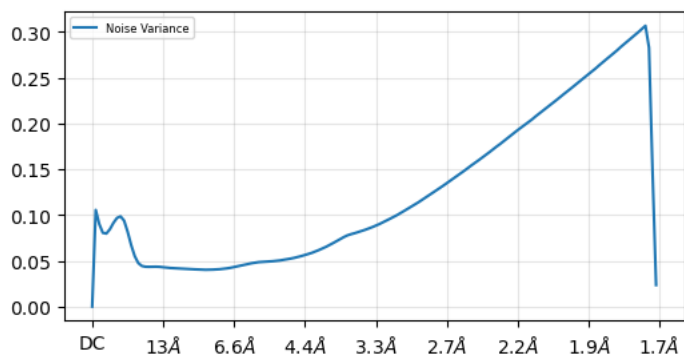
> [2026-06-09 15:25:14.43 GMT-4] cFSCs (Half-angle: 20°) Iteration 002, with tight mask



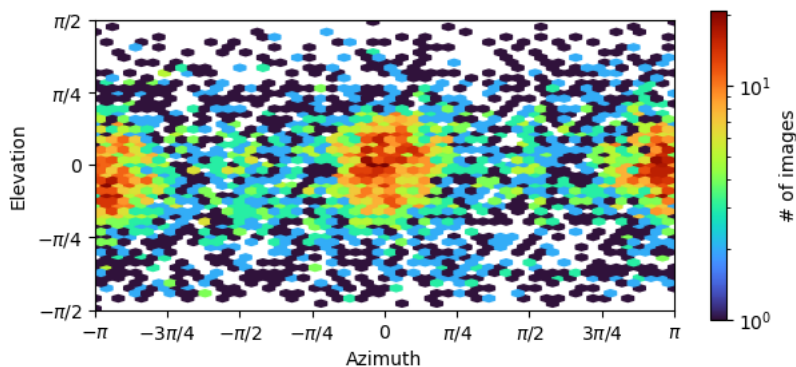
> [2026-06-09 15:25:20.76 GMT-4] Guinier Plot Iteration 002



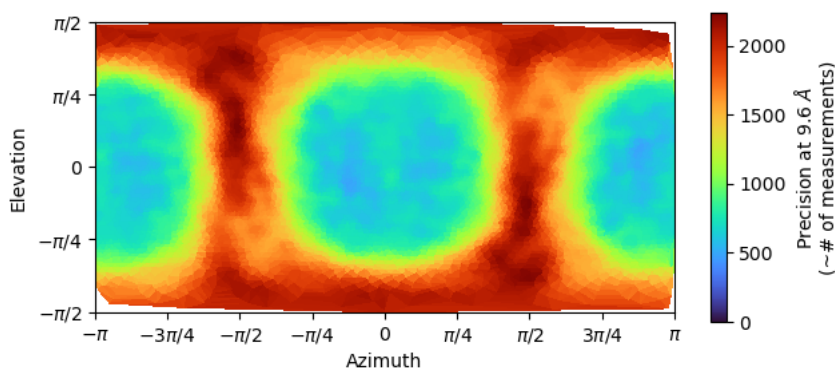
> [2026-06-09 15:25:21.12 GMT-4] Noise Model Iteration 002



> [2026-06-09 15:25:21.88 GMT-4] Viewing Direction Distribution Iteration 002



> [2026-06-09 15:25:23.51 GMT-4] Posterior Precision Directional Distribution Iteration 002



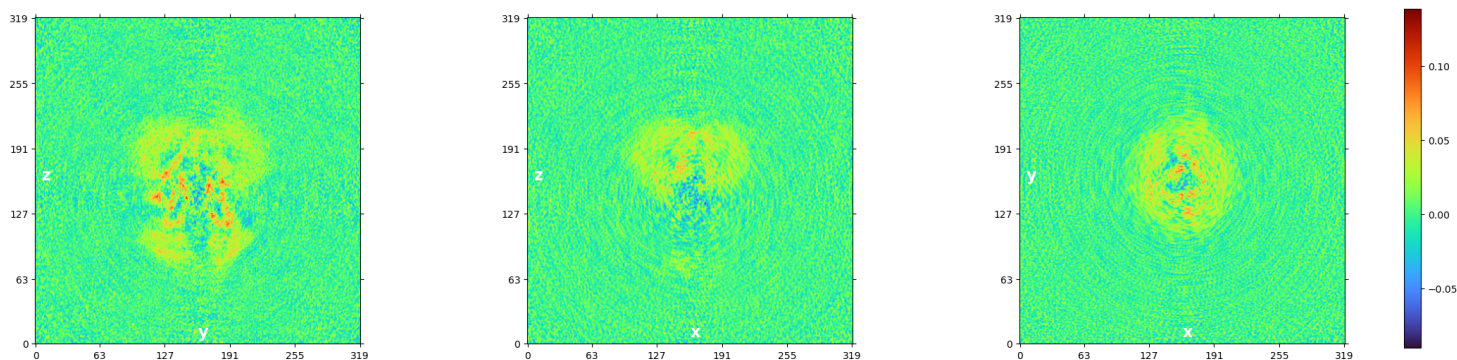
```
> [2026-06-09 15:25:23.51 GMT-4] [CPU: 10.75 GB] [Avail: 290.05 GB] Done in 21.300s.
> [2026-06-09 15:25:23.52 GMT-4] [CPU: 10.75 GB] [Avail: 290.05 GB] Outputting files..
> [2026-06-09 15:25:40.20 GMT-4] [CPU: 10.65 GB] [Avail: 289.73 GB] Done in 16.679s.
> [2026-06-09 15:25:40.20 GMT-4] [CPU: 10.65 GB] [Avail: 289.73 GB] Done iteration 2 in 166.730s. Total time so far 501.816s
> [2026-06-09 15:25:41.16 GMT-4] [CPU: 10.65 GB] [Avail: 289.87 GB] ----- Start Iteration 3
> [2026-06-09 15:25:41.16 GMT-4] [CPU: 10.65 GB] [Avail: 289.86 GB] Using Max Alignment Radius 34.069 (7.796A)
> [2026-06-09 15:25:41.17 GMT-4] [CPU: 10.65 GB] [Avail: 289.86 GB] Using Full Dataset (split 2747 in A, 2731 in B)
> [2026-06-09 15:25:41.17 GMT-4] [CPU: 10.65 GB] [Avail: 289.86 GB] Using dynamic mask.
> [2026-06-09 15:26:12.31 GMT-4] [CPU: 10.65 GB] [Avail: 291.35 GB] -- THR 1 BATCH 500 NUM 866 TOTAL 2.5010082 ELAPSED 20.280230
--
> [2026-06-09 15:26:35.21 GMT-4] [CPU: 11.41 GB] [Avail: 291.75 GB] Processed 5478.000 images in 22.894s.
> [2026-06-09 15:26:39.09 GMT-4] [CPU: 11.41 GB] [Avail: 291.47 GB] Computing FSCs...
> [2026-06-09 15:26:39.09 GMT-4] [CPU: 11.41 GB] [Avail: 291.47 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:26:39.09 GMT-4] [CPU: 11.41 GB] [Avail: 291.47 GB] Computing FFTs on GPU.
> [2026-06-09 15:26:42.64 GMT-4] [CPU: 11.41 GB] [Avail: 292.20 GB] Done in 3.551s
> [2026-06-09 15:26:42.66 GMT-4] [CPU: 11.41 GB] [Avail: 292.41 GB] Optimizing FSC Mask...
> [2026-06-09 15:26:42.66 GMT-4] [CPU: 11.41 GB] [Avail: 292.45 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:26:42.67 GMT-4] [CPU: 11.41 GB] [Avail: 292.45 GB] Computing FFTs on GPU.
> [2026-06-09 15:26:44.51 GMT-4] [CPU: 11.66 GB] [Avail: 292.06 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:26:44.51 GMT-4] [CPU: 11.66 GB] [Avail: 292.06 GB] Computing FFTs on GPU.
> [2026-06-09 15:26:46.58 GMT-4] [CPU: 11.66 GB] [Avail: 291.85 GB] Using full box size 320, downsampled box size 160, with low
```



```

> [2026-06-09 15:26:46.59 GMT-4] [CPU: 11.66 GB] [Avail: 291.85 GB] Computing FFTs on GPU.
> [2026-06-09 15:26:48.59 GMT-4] [CPU: 11.66 GB] [Avail: 291.47 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:26:48.60 GMT-4] [CPU: 11.66 GB] [Avail: 291.47 GB] Computing FFTs on GPU.
> [2026-06-09 15:26:50.75 GMT-4] [CPU: 11.66 GB] [Avail: 292.27 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:26:50.75 GMT-4] [CPU: 11.66 GB] [Avail: 292.27 GB] Computing FFTs on GPU.
> [2026-06-09 15:26:52.62 GMT-4] [CPU: 11.66 GB] [Avail: 292.08 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:26:52.63 GMT-4] [CPU: 11.66 GB] [Avail: 292.12 GB] Computing FFTs on GPU.
> [2026-06-09 15:26:54.64 GMT-4] [CPU: 11.66 GB] [Avail: 292.19 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:26:54.65 GMT-4] [CPU: 11.66 GB] [Avail: 292.18 GB] Computing FFTs on GPU.
> [2026-06-09 15:26:56.76 GMT-4] [CPU: 11.66 GB] [Avail: 292.22 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:26:56.77 GMT-4] [CPU: 11.66 GB] [Avail: 292.22 GB] Computing FFTs on GPU.
> [2026-06-09 15:26:58.75 GMT-4] [CPU: 11.66 GB] [Avail: 290.52 GB] Done in 16.088s
> [2026-06-09 15:26:58.76 GMT-4] [CPU: 11.66 GB] [Avail: 290.50 GB] Computing cFSCs...
> [2026-06-09 15:27:00.84 GMT-4] [CPU: 11.66 GB] [Avail: 291.49 GB] Done in 2.078s
> [2026-06-09 15:27:00.84 GMT-4] [CPU: 11.66 GB] [Avail: 291.49 GB] Using Filter Radius 33.793 (7.860A) | Previous: 34.069
(7.796A)
> [2026-06-09 15:27:06.88 GMT-4] [CPU: 11.03 GB] [Avail: 290.81 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:27:06.91 GMT-4] [CPU: 10.91 GB] [Avail: 290.87 GB] Running local cross validation for A ...
> [2026-06-09 15:27:29.39 GMT-4] [CPU: 11.16 GB] [Avail: 290.85 GB] Local cross validation A done in 22.480s
> [2026-06-09 15:27:30.65 GMT-4] FSC Filtered Side A

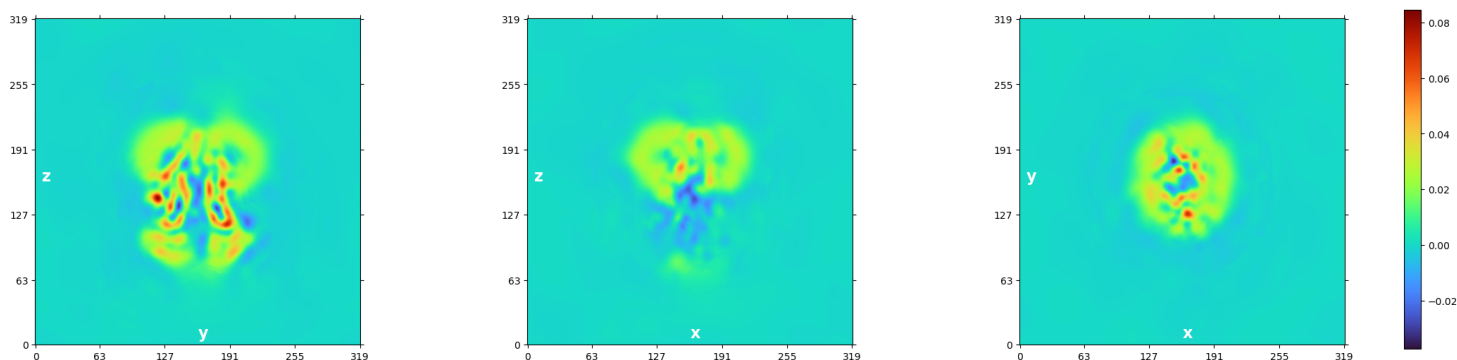
```



```

> [2026-06-09 15:27:31.87 GMT-4] CV Filtered Side A

```

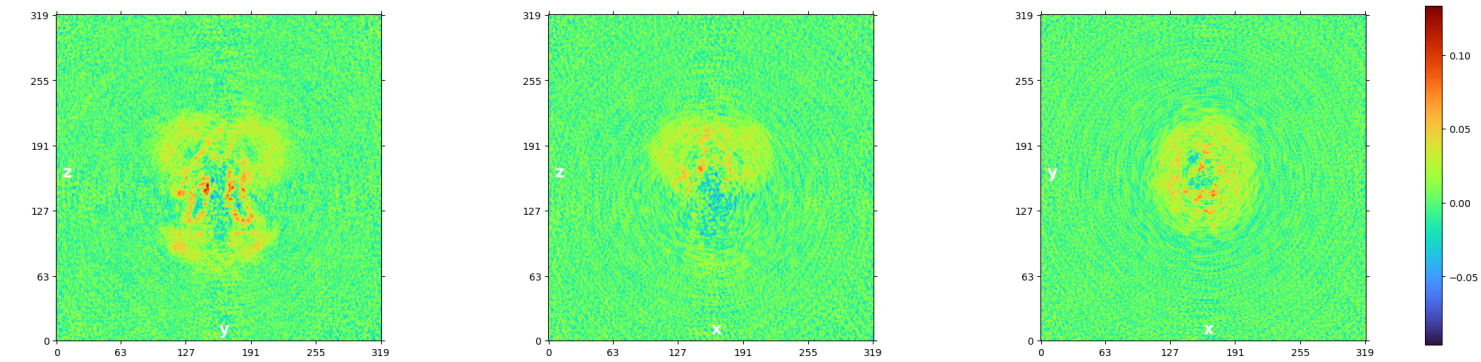


```

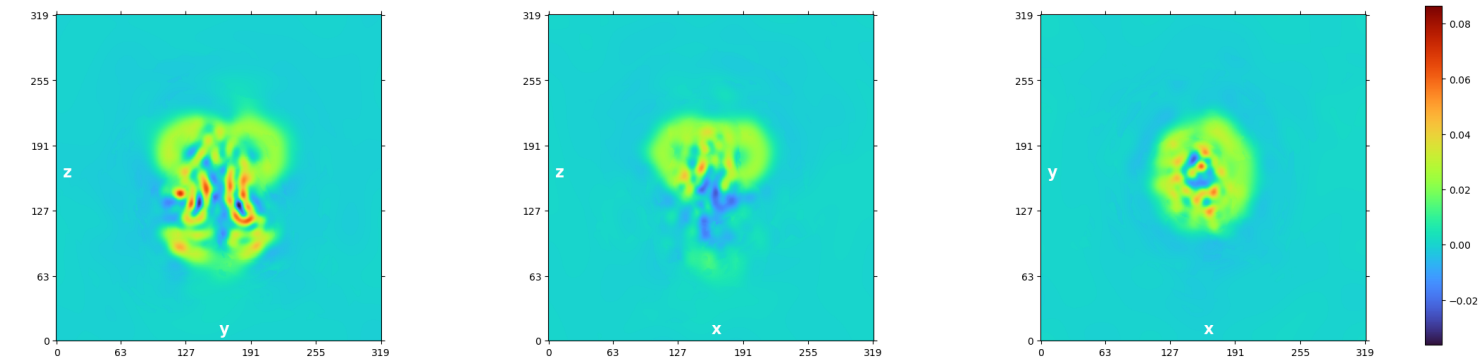
> [2026-06-09 15:27:31.87 GMT-4] [CPU: 11.16 GB] [Avail: 292.63 GB] Running local cross validation for B ...
> [2026-06-09 15:27:54.58 GMT-4] [CPU: 11.41 GB] [Avail: 291.11 GB] Local cross validation B done in 22.709s

```

> [2026-06-09 15:27:56.09 GMT-4] FSC Filtered Side B



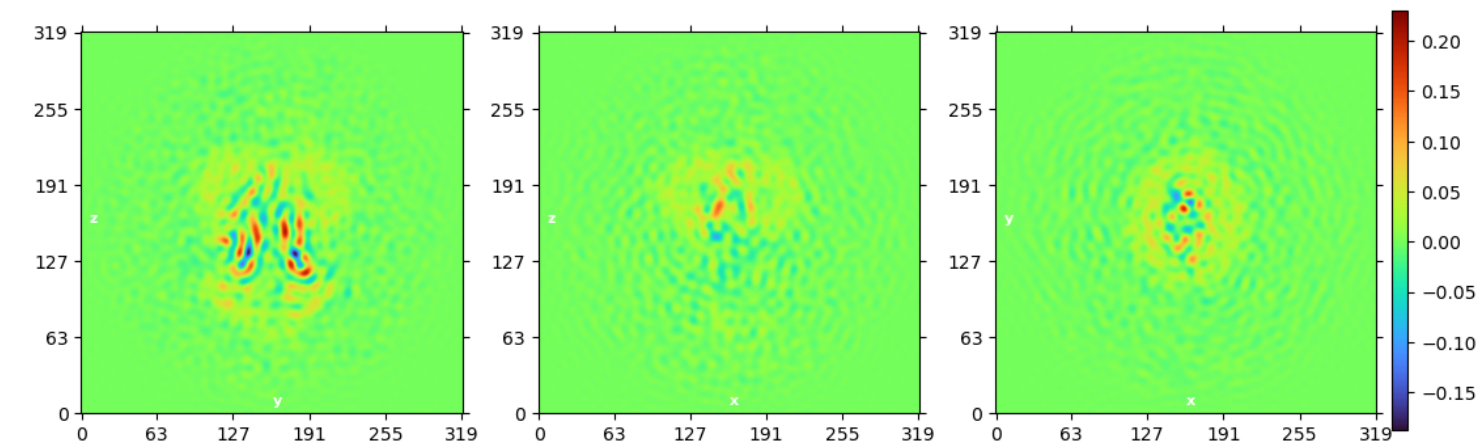
> [2026-06-09 15:27:57.20 GMT-4] CV Filtered Side B



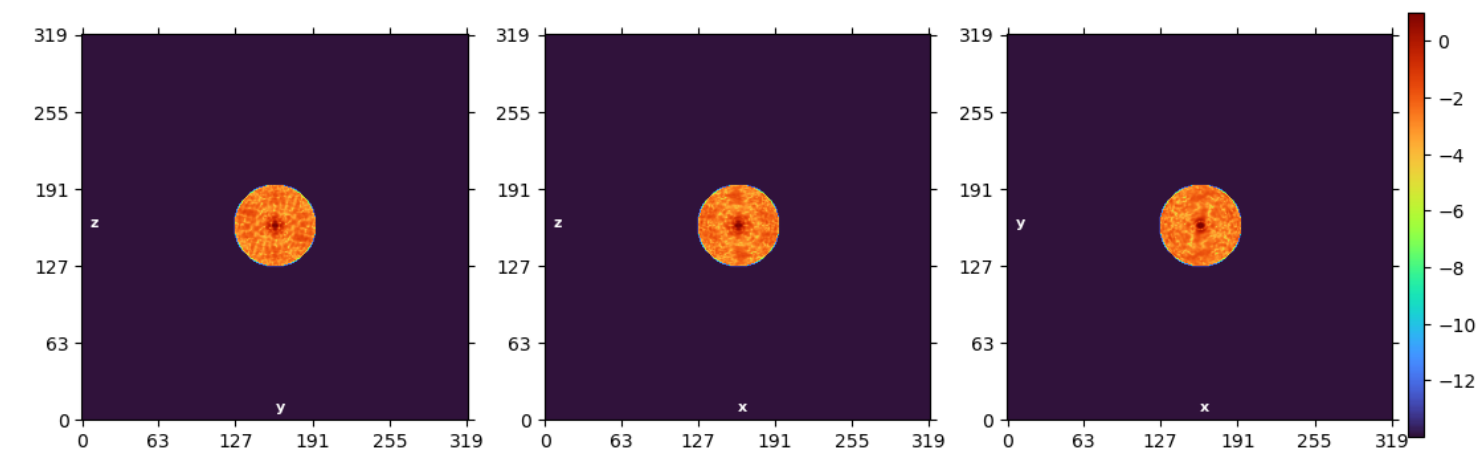
> [2026-06-09 15:28:04.84 GMT-4] [CPU: 11.28 GB] [Avail: 291.41 GB] Estimated Bfactor: -715.4

> [2026-06-09 15:28:04.84 GMT-4] [CPU: 11.28 GB] [Avail: 291.41 GB] Plotting..

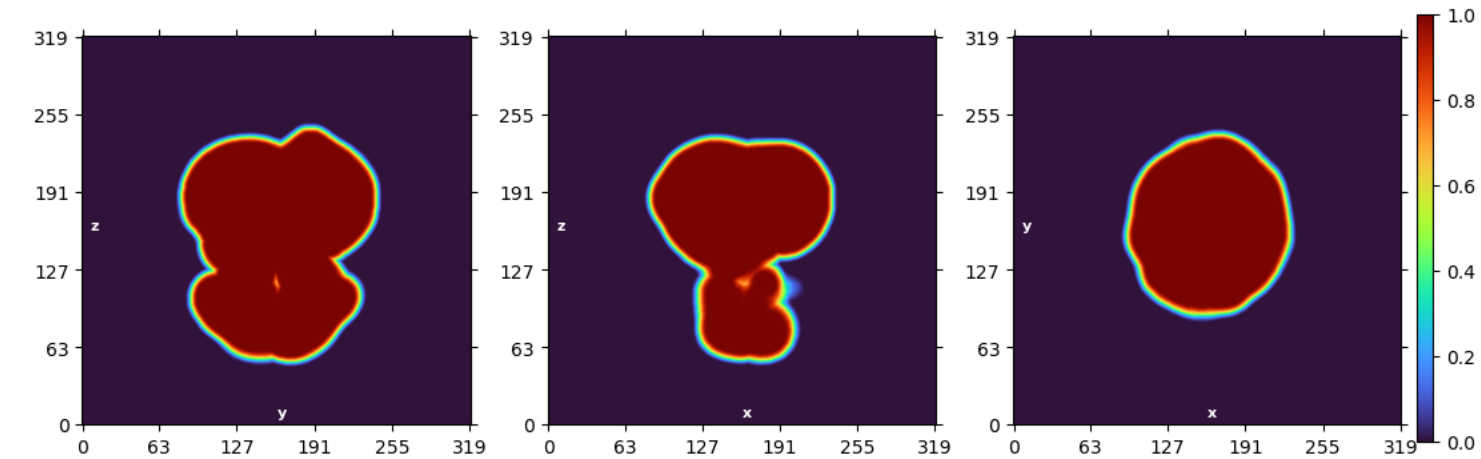
> [2026-06-09 15:28:12.07 GMT-4] Real Space Slices Iteration 003



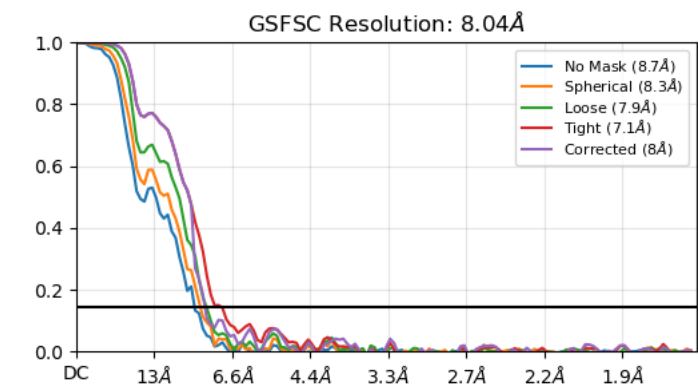
> [2026-06-09 15:28:13.58 GMT-4] Fourier Space Slices Iteration 003



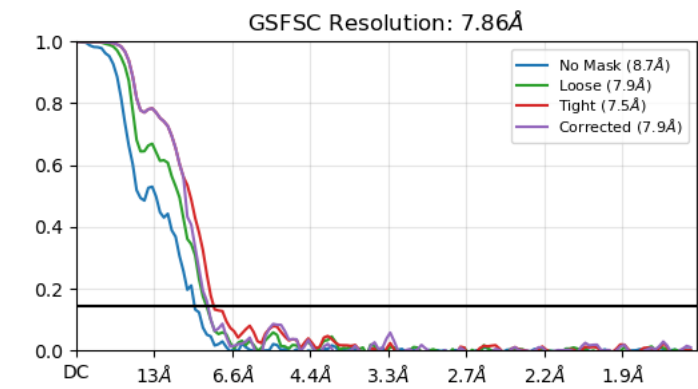
> [2026-06-09 15:28:15.85 GMT-4] Real Space Mask Slices Iteration 003



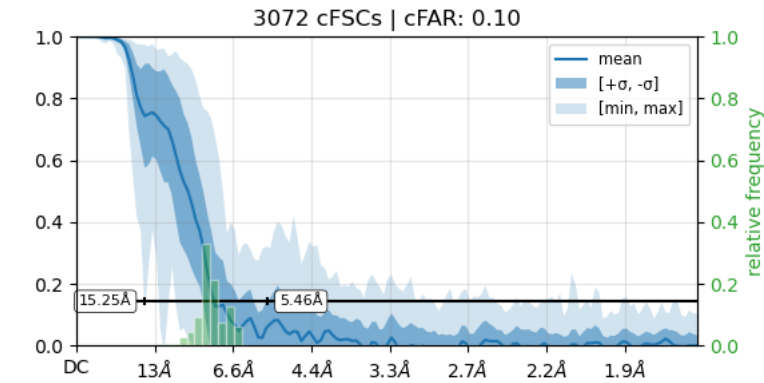
> [2026-06-09 15:28:16.50 GMT-4] FSC Iteration 003



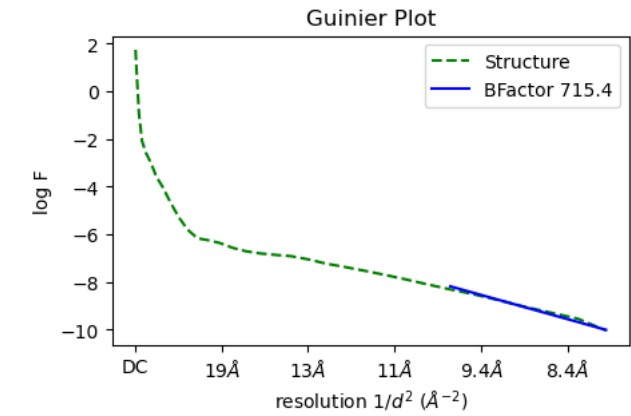
> [2026-06-09 15:28:17.10 GMT-4] FSC Iteration 003, after FSC-mask auto-tightening



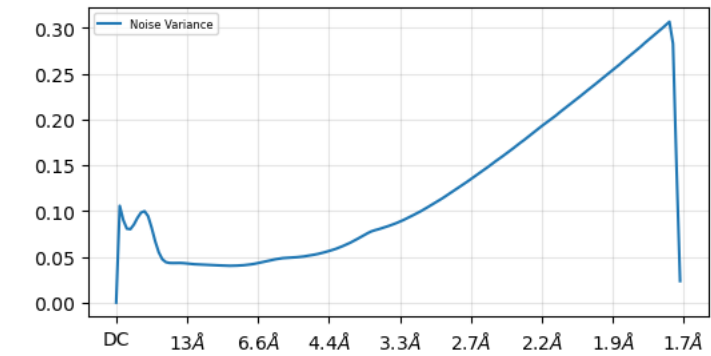
> [2026-06-09 15:28:17.96 GMT-4] cFSCs (Half-angle: 20°) Iteration 003, with autotight mask



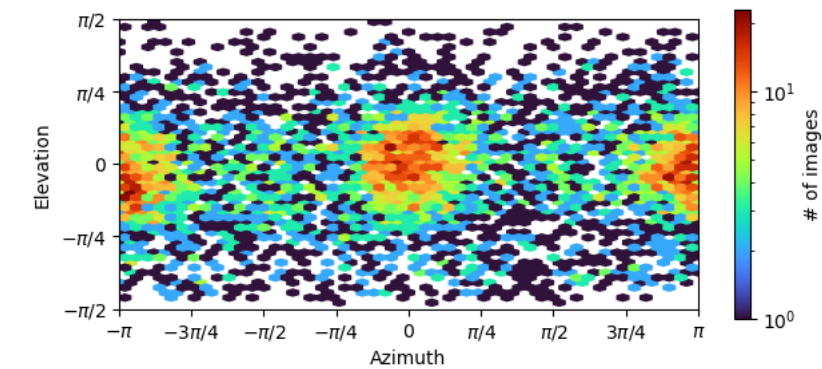
> [2026-06-09 15:28:23.67 GMT-4] Guinier Plot Iteration 003



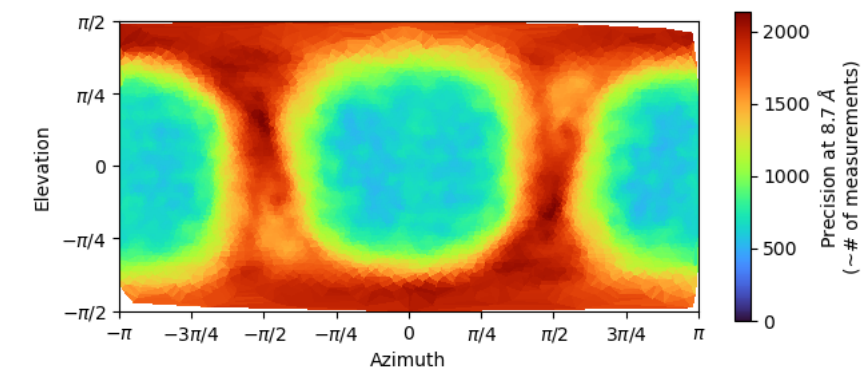
> [2026-06-09 15:28:24.04 GMT-4] Noise Model Iteration 003



> [2026-06-09 15:28:24.80 GMT-4] Viewing Direction Distribution Iteration 003



> [2026-06-09 15:28:26.28 GMT-4] Posterior Precision Directional Distribution Iteration 003



> [2026-06-09 15:28:26.29 GMT-4] [CPU: 11.34 GB] [Avail: 290.43 GB] Done in 21.439s.

> [2026-06-09 15:28:26.29 GMT-4] [CPU: 11.34 GB] [Avail: 290.43 GB] Outputting files..

> [2026-06-09 15:28:43.46 GMT-4] [CPU: 11.25 GB] [Avail: 290.06 GB] Done in 17.163s.

> [2026-06-09 15:28:43.46 GMT-4] [CPU: 11.25 GB] [Avail: 290.09 GB] Done iteration 3 in 182.295s. Total time so far 685.073s

> [2026-06-09 15:28:44.19 GMT-4] [CPU: 11.25 GB] [Avail: 291.02 GB] ===== Done Refinement =====

> [2026-06-09 15:28:44.19 GMT-4] [CPU: 11.25 GB] [Avail: 291.05 GB] Note that the output structure from refinement has been lowpass filtered to the gold-standard FSC resolution estimated above. However, the structure probably needs sharpening in order to best visualize high resolution features. This can be


```

done with the Sharpening task (as a new experiment).
> [2026-06-09 15:28:44.19 GMT-4] [CPU: 11.25 GB] [Avail: 291.06 GB] Full run took 685.810s
> [2026-06-09 15:28:44.87 GMT-4] [CPU: 5.39 GB] [Avail: 296.31 GB] -----
-
> [2026-06-09 15:28:44.87 GMT-4] [CPU: 5.39 GB] [Avail: 296.30 GB] Compiling job outputs...
> [2026-06-09 15:28:44.88 GMT-4] [CPU: 5.39 GB] [Avail: 296.30 GB] Passing through outputs for output group particles from input
group particles
> [2026-06-09 15:28:44.90 GMT-4] [CPU: 5.39 GB] [Avail: 296.36 GB] This job outputted results ['alignments3D', 'ctf']
> [2026-06-09 15:28:44.91 GMT-4] [CPU: 5.39 GB] [Avail: 296.34 GB]   Loaded output dset with 5478 items
> [2026-06-09 15:28:44.91 GMT-4] [CPU: 5.39 GB] [Avail: 296.33 GB] Passthrough results ['blob', 'alignments3D_multi']
> [2026-06-09 15:28:44.99 GMT-4] [CPU: 5.39 GB] [Avail: 296.08 GB]   Loaded passthrough dset with 5478 items
> [2026-06-09 15:28:45.02 GMT-4] [CPU: 5.39 GB] [Avail: 296.02 GB]   Intersection of output and passthrough has 5478 items
> [2026-06-09 15:28:45.04 GMT-4] [CPU: 5.39 GB] [Avail: 296.01 GB]   Output dataset contains: ['blob', 'alignments3D_multi']
> [2026-06-09 15:28:45.04 GMT-4] [CPU: 5.39 GB] [Avail: 296.01 GB]   Outputting passthrough result blob
> [2026-06-09 15:28:45.05 GMT-4] [CPU: 5.39 GB] [Avail: 296.00 GB]   Outputting passthrough result alignments3D_multi
> [2026-06-09 15:28:45.06 GMT-4] [CPU: 5.39 GB] [Avail: 295.99 GB] Checking outputs for output group particles
> [2026-06-09 15:28:45.14 GMT-4] [CPU: 5.40 GB] [Avail: 295.79 GB] Updating job size...
> [2026-06-09 15:28:45.42 GMT-4] [CPU: 5.40 GB] [Avail: 295.38 GB] Exporting job and creating csg files...
> [2026-06-09 15:28:45.97 GMT-4] [CPU: 5.40 GB] [Avail: 295.02 GB] *****
**
> [2026-06-09 15:28:45.97 GMT-4] [CPU: 5.40 GB] [Avail: 295.01 GB] Job complete. Total time 687.62s

```